Disciplinary and Interdisciplinary Science Education Research

REVIEW

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Scientific practices in science education, 2013–2023: bibliometric mapping and thematic synthesis

Yaling Chi[1,2], Lei Wang[1,3*] and Xijuan Li[1]

Abstract

As a core dimension of science education, scientific practices have shifted the field's focus from knowledge transmission to engaging students in authentic inquiry and problem-solving, and this shift has been accompanied by a growing body of related scholarship. This review integrates CiteSpace-based bibliometric visualization with manual thematic content analysis to map international research on scientific practices (2013–2023) and characterize its development. Publication output increased over 2013–2023 and was predominantly concentrated at the secondary and undergraduate levels, while early childhood and cross-grade contexts remain underrepresented. Collaboration networks are highly centered on U.S. institutions, suggesting opportunities to foster more geographically and culturally diverse partnerships. Keyword clustering highlights teachers, students, and tasks as focal dimensions of practice-oriented inquiry. Temporal evidence from the timeline view and burst detection suggests shifting emphases from conceptual framing to instructional enactment, then to assessment-oriented refinement and instructional contextualization, with more recent attention to technology-mediated enactment—indicating cumulative layering rather than discrete transitions. At the research-design level, emerging alignments are evident among research questions, methods, and theoretical or analytical frameworks, yet uneven coupling persists in specific pairings (e.g. assessment–learning progressions; instrument development–inquiry-oriented frameworks). These mapped patterns suggest three implications: (a) extend research to underrepresented populations and strengthen cross-regional collaboration to enable contextual comparison; (b) use integrated bibliometric–thematic workflows to triangulate interpreted theme shifts with content-coded evidence and improve interpretive transparency; and (c) tighten design–measurement coherence in weakly coupled areas by aligning constructs, tasks, and validity arguments from the outset.

Keywords Scientific practices, Science education, Bibliometric analysis, Content analysis, Research topic evolution

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Introduction

Amid rapid technological innovation, the rapid expansion of knowledge, and deepening globalization, school science education increasingly relies on evidence-based reasoning and authentic, context-rich problem solving to cultivate future-oriented scientific literacy. More than a decade ago, the Framework for K–12 Science Education (National Research Council, 2012) and the Next Generation Science Standards (NGSS Lead States, 2013) jointly established science and engineering practices as a core dimension of three-dimensional learning–alongside cross-cutting concepts and disciplinary core ideas. This policy shift not only foregrounded the principle of learning through practice but also catalyzed what some scholars describe as a broader "practice turn" in science education–a shift from treating learning primarily as the acquisition of decontextualized knowledge to emphasizing students' authentic participation in socioculturally situated disciplinary practices (Ford & Forman, 2006; Forman, 2018; Kelly, 2014).

From a sociocultural perspective, science learning is not merely a matter of individual knowledge acquisition; rather, it is a process through which learners construct meaning and identities via social collaboration, the mediated use of tools, and discursive practices within specific communities of practice Kelly (2014); Cole (1996). Consistent with this view, the three-dimensional learning framework integrates scientific practices, cross-cutting concepts, and disciplinary core ideas, thereby reconfiguring how students develop scientific understanding (National Research Council, 2012; NGSS Lead States, 2013). Against this backdrop, the literature on scientific practices has expanded rapidly, encompassing instructional models, assessment systems, curriculum design, and students' identity development (e.g., Bybee, 2014; Laverty et al., 2016; Costa et al., 2021; Avraamidou, 2020), and increasingly linking theoretical advances with empirical inquiry. Given the growth of this body of work, an integrative review is needed to map the field's knowledge structure and evolving trajectories in terms of publication patterns， thematic emphases, and research designs, thereby informing future research agendas and methodological innovation.

From an international perspective, many countries and regions have incorporated practice-oriented approaches into science curriculum policy, albeit in different ways. England's National Curriculum foregrounds "working scientifically" as a means of connecting inquiry, evidence, and communication. Australia's curriculum, developed by the Australian Curriculum, Assessment and Reporting Authority (ACARA), specifies science inquiry skills, emphasizing question posing, evidence use, and communication. Singapore's science curriculum organizes practical work around inquiry as a unifying thread. Finland's National Core Curriculum highlights phenomenon-based, cross-disciplinary learning grounded in inquiry. China's recently promulgated curriculum standards and its core competencies framework identify scientific inquiry, evidence-based reasoning, and model-based representation as key dimensions of student competence. Despite variations in terminology, these systems share a common logic: treating practice as a central route to cultivating scientific literacy (Department for Education, 2013; Australian Curriculum, Assessment and Reporting Authority [ACARA], n.d; Ministry of Education, Singapore, 2013, Finnish National Board of Education, 2014; Ministry of Education of the People's Republic of China, 2022).

Despite notable growth, existing reviews in science education exhibit several recurrent limitations. First, many syntheses rely on older or small-scale corpora, making it difficult to capture the field's recent rapid expansion as well as the emergence of new hotspots and shifts in research foci (Costa & Broietti, 2021; Leowardy et al., 2025; Rönnebeck et al., 2016). Second, traditional systematic reviews still predominate; although they apply rigorous search and quality appraisal procedures, they are time-consuming, heavily reviewer-dependent, and often constrained in timeliness and breadth of coverage, rendering them less well suited for tracing the field's macro-level structure and evolution in a fast-growing domain Petticrew and Roberts (2006); Gough et al. (2012). Third, extant reviews rarely offer a systematic examination of the alignment among research questions, methodological choices, and theoretical or analytical frameworks. Fourth, from an international and cross-cultural perspective, most reviews remain largely descriptive–reporting counts by region or educational level–while comparative analysis using structural indicators (e.g., centrality and cohesion) remain limited (Tsai & Wen, 2005; Lee et al., 2009).

In addition to broad comprehensive or scoping reviews of scientific practices, a growing number of subdomain-focused secondary studies have also emerged. However, these studies often concentrate on a single practice, a particular region or grade band, or an earlier time period, which limits their ability to offer a comprehensive, cross-practice and cross-context perspective on developments over the past decade (Jin & Jian, 2024; Jian et al., 2023; Tho et al., 2017; Sun et al., 2021; Tikva & Tambouris, 2021; Hendratmoko et al., 2024; Nicolaou & Constantinou, 2014).

To address these limitations, this study develops and employs an integrative mixed-methods design that combines CiteSpace-based bibliometric visualization with manual thematic analysis. On the bibliometric side, algorithmic co-occurrence and co-citation analyses, together with burst detection and network visualization, are used

to systematically reveal the field's macro-level knowledge structure and the evolution of research frontiers over time (Chen, 2006; Chen et al., 2010; Zupic & Čater, 2015). On the qualitative side, thematic content analysis operates at the micro-level to elucidate the meanings of core constructs, their relevance across different practice settings, and the internal logic linking research questions, methodological choices, and theoretical or analytical frameworks. These two strands are mutually reinforcing: together, they mitigate subjective-related bias while preserving deep, context-sensitive explanatory power (Gough et al., 2012; Petticrew & Roberts, 2006).

In this article, scientific practices are defined with reference to the practice repertoire articulated in the Framework for K–12 Science Education and the Next Generation Science Standards (National Research Council, 2012; NGSS Lead States, 2013). Our analysis focuses on how these practices are enacted–and with what effects–across key sites of curriculum, instruction, assessment, and teacher professional development. To enhance operational and analytical clarity, we further operationalize scientific practices as the eight NGSS-defined practice categories: asking questions and defining problems; developing and using models; planning and carrying out investigations; analyzing and interpreting data; using mathematics and computational thinking; constructing explanations and designing solutions; engaging in argument from evidence; and obtaining, evaluating, and communicating information (National Research Council, 2012; NGSS Lead States, 2013). Accordingly, we foreground scientific practices as an integrated pathway for advancing students' scientific literacy rather than isolating any single practice for standalone treatment.

We systematically searched for and analyzed prior review articles on scientific practices within the scope of this study and identified recurring gaps. Costa and Broietti (2021) surveyed 44 publications (2010–2019), primarily categorizing and describing research contexts. However, they did not fully leverage bibliometric visualization to map collaboration networks, keyword clusters and burst patterns, or co-citation structures, nor did they systematically examine alignment among research questions, methods, and theoretical or analytical frameworks. In a scoping review, Leowardy et al. (2025) expanded the time window to 2011–2021 and included 229 studies, thereby offering broader thematic coverage. Nonetheless, post-2021 developments were beyond the scope of their review window, and they likewise did not provide an integrated analysis of the research question–method–framework triad. Taken together, existing reviews still leave room for a more integrated account of linkages among structure, themes, and research design over the past decade, limiting cross-context comparability and longitudinal trend assessment (Costa & Broietti, 2021; Leowardy et al., 2025). Accordingly, the present study combines bibliometric analysis with qualitative content/thematic analysis to (a) map research outputs and collaboration networks, (b) identify thematic clusters and hotspot dynamics, and (c) examine–at the level of research design – correspondence patterns and underlying tensions among research questions, methods, and theoretical or analytical frameworks.

This study contributes to research on scientific practices in science education in three ways. First, it systematically maps the field by educational level and discipline context, and analyzes the degree of centralization and the structure of cross-national and cross-institutional collaboration. Second, it examines the field's major thematic areas and tracks their evolution over time, focusing on how emphasis may shift across conceptual, instructional, assessment-oriented, and technology-enhanced lines of inquiry. Third, it investigates patterns of alignment–and potential tensions–among research questions, methods, and theoretical/analytical frameworks (e.g., assessment vs. learning progressions; instrument development vs. inquiry frameworks), and proposes a transferable analytic approach. The research questions are as follows:

RQ 1. What does bibliometric analysis reveal about the productivity and collaboration landscape of research on scientific practices in science education, including annual publication trends, geographic and institutional distributions, and the structure of international collaboration networks?

RQ 2. What does manual content analysis reveal about the distributions of this research across educational levels and disciplinary contexts?

RQ 3. Based on keyword clustering, what major research themes can be identified, and how have research hotspots evolved from 2013 to 2023?

RQ 4. At the level of research content, what types of research questions are most prevalent? Which methods and theoretical/analytical frameworks are commonly employed? Do specific recurring alignment patterns emerge among these three dimensions?

Methodology and analytical procedures

This study combines bibliometric mapping and thematic content analysis to systematically characterize the knowledge base, structural features, and evolutionary trajectory of research on scientific practices in science education, and to derive empirically grounded directions for future work. The bibliometric component leverages bibliographic metadata and network analysis to examine annual publication trends and the structure of institutional collaboration networks. It also uses keyword co-occurrence analysis and timeline-based clustering to trace theme evolution and changes in research hotspots

(Zupic & Čater, 2015; Pinto et al., 2019). Dimensions that are not reliably captured in database metadata—such as educational level, school subject, and participant characteristics—are manually coded using a thematic content analysis scheme. This mixed approach preserves a macro-level overview while ensuring accurate classification of sampled studies and their disciplinary contexts relevant to our research questions.

Building on this mapping, we further conduct a descriptive content analysis. Following prior typologies (Çalık & Sözbilir, 2014; Dinçer, 2018), we distinguish among meta-analysis, meta-synthesis (thematic synthesis), and descriptive content analysis, and we adopt the latter to construct and examine a three-way cross-tabulation of research questions × methods × theoretical/analytical frameworks (RQ × M × F). We compute adjusted standardized residuals (ASRs) for each cell in the contingency table to identify statistically meaningful over- and under-represented combinations, thereby informing future theory building and methodological integration (Walsh & Downe, 2005).

To ensure transparency and reproducibility, we adopted a study-specific workflow for literature identification and screening, documenting the key steps and criteria in this section. Given that this study is a bibliometric–thematic review, we report only information relevant to searching and screening; risk-of-bias assessment and effect-size synthesis are beyond the scope of this work. Data were retrieved from Clarivate's Web of Science Core Collection (WoSCC), which provides broad coverage and citation-linking functions that facilitate corpus definition and reproducible downstream analyses (Mongeon & Paul-Hus, 2016; Li et al., 2018). As shown in Fig. 1, we retrieved 406 records, removed duplicates ($n = 2$), screened titles and abstracts for relevance, and excluded out-of-scope items ($n = 141$), yielding a final corpus of 263 studies for bibliometric mapping and thematic content analysis. The subsections that follow detail the data sources, search strategy, and analytical procedures.

Data collection

Data collection process for bibliometric analysis

A core objective of this study is to apply bibliometric methods to analyze publication patterns and to trace the evolving trajectories of research on scientific practices in science education. As the literature has expanded rapidly in recent years, it has become increasingly challenging for researchers to maintain a comprehensive view of the field's shifting landscape. In this context, bibliometric approaches are well suited to processing large-scale corpora, identifying influential contributions through citation-based indicators, and generating structured, reproducible evidence to support field-level synthesis (Zupic & Čater, 2015). By statistically analyzing bibliographic metadata–such as authorship, keywords, citations, and publication venues–bibliometrics can reveal patterns of scholarly communication and delineate the

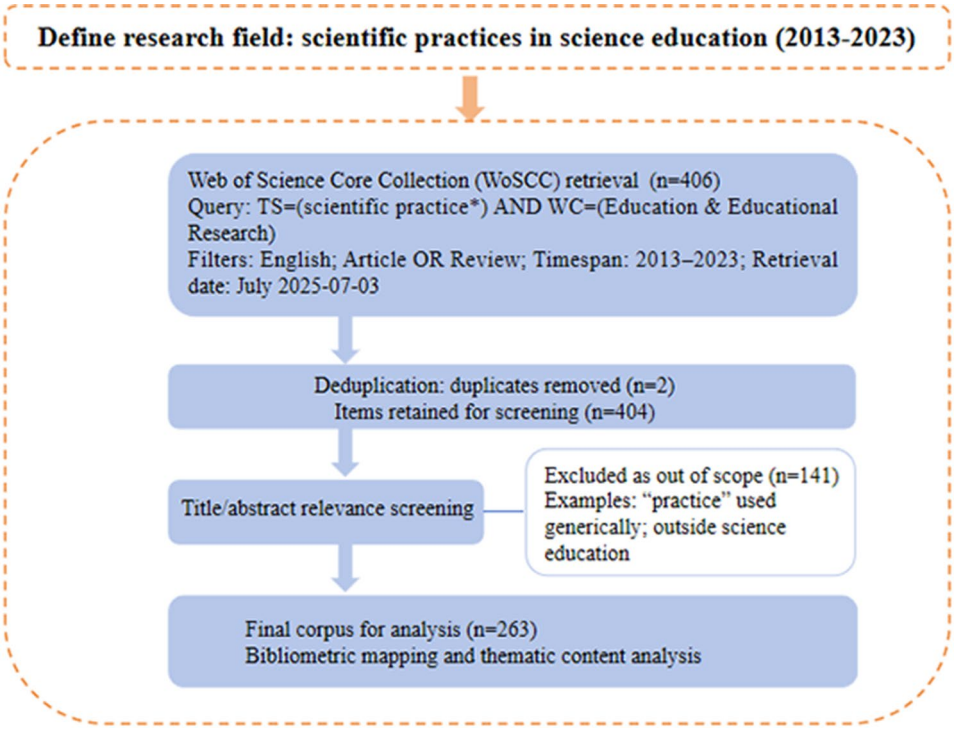


Fig. 1 Study-specific workflow for data retrieval and screening in this bibliometric–thematic review

knowledge structure and temporal dynamics of the field (Al & Tonta, 2004).

During data collection for the bibliometric analysis, we searched Clarivate's Web of WoSCC on 3 July 2025. The core query term was "scientific practice*", and retrieval was restricted to the Web of Science category "Education & Educational Research". We searched the Topic field (TS) using the query TS=("scientific practice*") AND WC = ("Education & Educational Research"), with the publication period restricted to July 2013 to July 2023. We further limited results to English-language records and document types Article and Review. This strategy was designed for a term-explicit, precision-oriented mapping of the use of "scientific practice(s)" in core educational research, rather than for exhaustive coverage of all practice-adjacent or SEP-related literatures. The initial search returned 406 records.

All analyses were conducted on a fixed snapshot exported on 3 July 2025 (export date: 2025–07-03) in the format "Plain Text–Full Record and Cited References". Records were deduplicated using DOI matching and title-based comparison, with manual verification for ambiguous cases. We then screened titles and abstracts for topical relevance, excluding items in which "practice" referred to general practice or training rather than the construct of scientific practices in science education; full-text checks were conducted when eligibility could not be determined from the title/abstract alone. The final analytic sample included 263 records. The literature search and screening workflow is summarized in Fig. 1.

Data analysis process for content analysis

The content analysis sample was identical to the bibliometric corpus ($n = 263$). No secondary searches or additional exclusions were performed, and the inclusion criteria were the same as those described in Sect. Data collection process for bibliometric analysis. To support the subsequent coding and reliability assessment, we pre-extracted and organized the following fields for each record: grade level, discipline, research questions (RQs), methodological orientation (M), and theoretical/analytical framework (F). Primary-level (L1) coding used single-label assignment, whereas subcodes at L2 permitted multi-label assignment (see Appendix S2). All entries were independently verified by two researchers against the source articles, and discrepancies were resolved through discussion to reach consensus. Intercoder agreement was assessed on a random 20% subsample ($n = 53$). Unweighted Cohen's κ indicated substantial to near-perfect agreement: grade level ($\kappa = 0.85$), discipline ($\kappa = 0.83$), RQs ($\kappa = 0.84$), methodological orientation ($\kappa = 0.81$), and framework ($\kappa = 0.79$). Corresponding 95% confidence intervals are reported in Appendix S2. For multi-label L2 subcodes, κ was computed per binary subcode and then macro-averaged. Accordingly, the corpus construction workflow is summarized in Fig. 1.

Data analysis

Data analysis process for bibliometric analysis

We used the "Full Record and Cited References" exports downloaded from Web of Science and conducted all bibliometric analyses on the same corpus ($n = 263$). First, using the "Publication Year" and "Times Cited" fields, we computed annual publication counts and extracted citation-impact indicators to depict the field's output trajectory and citation influence over time. Second, after reviewing each record's title, abstract, and author keywords, we manually coded the educational level, school subject, and participant group and then compiled frequency summaries to characterize distributions across these three dimensions (see Appendix S1 for the coding scheme). For transparency, Web of Science served only as the source of raw records; we did not use the platform's built-in analytics functions.

We subsequently used CiteSpace (version 6.3.R3) to construct collaboration networks and characterize the global collaboration landscape and central actors. Specifically, we conducted separate runs using countries and institutions as node types. We set the time slicing to 1 year and used cosine similarity to measure inter-node relatedness. Network pruning was performed using Pathfinder and Pruning sliced networks, with a Minimum Spanning Tree (MST) applied where appropriate. All other parameters were kept at their default settings.

To enhance readability, we applied a minimum occurrence threshold for node inclusion (adjusted to network scale) and removed isolated nodes; specific thresholds are reported alongside the corresponding results. In the final visualizations, node size is proportional to publication output, edge width reflects collaboration strength, and colors denote either cluster membership or time slices, as indicated in each figure. Together, these conventions provide an intuitive view of collaborating actors and the structural characteristics of the collaboration network.

At the thematic level, we treated keywords as nodes and conducted keyword co-occurrence, clustering, and timeline analyses in separate steps. We set the time slicing to 1 year and selected the top 15 most frequent terms per year (Top $N = 15$) for keyword selection. To improve representativeness and reduce noise, we normalized raw keywords by unifying capitalization and singular/plural forms, merging synonyms or near-synonyms, and removing generic, non-informative terms. Network construction used cosine similarity as the association measure, with Pathfinder and pruning sliced networks applied for sparsification. We then clustered the keyword co-occurrence network, with cluster labels automatically generated using latent semantic indexing (LSI). The

final outputs included both cluster and timeline views to visualize structural features and temporal dynamics. We report the overall modularity (Q) and mean silhouette (S) for major clusters; as a rule of thumb, $Q > 0.30$ and $S \geq 0.70$ suggest reasonably well-defined and cohesive clustering. Based on these outputs, we summarize the composition of major research themes and describe their trajectories across the study period.

Data analysis process for content analysis

Building on the bibliometric analysis, we conducted a descriptive content analysis on the same corpus ($n = 263$). While content analysis is widely used in qualitative research, it also enables systematic quantification of textual data, providing a useful bridge between qualitative interpretation and quantitative summary (Bauer, 2000). Following established guidance, we applied a clearly specified conceptual–thematic framework to categorize comparable units of meaning and reported the results in a transparent and auditable manner to surface underlying patterns in the literature (Creswell, 2012).

In the present study, we drew on the paper classification form (PCF) proposed by Çiltaş, Çiltaş et al. (2012) and adapted it to our aims to develop a study-specific coding framework. Using this framework, we systematically extracted three key variables from all included records: research question type (RQ type), methodological orientation, and theoretical/analytical framework. All records were entered into a Microsoft Excel–based article classification sheet and coded according to predefined rules; we then computed the frequencies and percentages for each category. We subsequently constructed a three-way contingency table (RQ types × M × F) and used adjusted standardized residuals (ASRs) to identify statistically noteworthy over- and under-represented combinations. Detailed coding rules and boundary-case examples are provided in Appendix S2.

Results

We report the results in the order of the research questions and analytical procedures outlined above. Consistent with our integrated bibliometric mapping and content-analytic approach, we first present a bibliometric overview, including trends in annual publications and citations as well as the institutional landscape of international collaboration. We then report keyword-based thematic mapping results, using co-occurrence networks, clustering, and timeline visualizations to delineate the thematic structure and its staged evolution. Finally, based on manual coding, we describe corpus characteristics (e.g., distributions by educational level, discipline, and participant group) and summarize the typologies and distributions of research question types, methods, and theoretical/analytical frameworks. We further report the cross-tabulations of RQ types × methods × theoretical/analytical frameworks, together with ASR heatmaps, to identify significant salient alignments and gaps. Unless otherwise noted, all analyses draw on the same corpus ($n = 263$).

Bibliometric overview of research on scientific practices in education

Annual distribution of publications and citations

Annual publication output and citation trends help reveal the evolutionary trajectory of a field and shifts in scholarly attention (Waltman, 2016; Wang et al., 2023). As shown in Fig. 2, the number of annual publications and citations in this area increased overall from 2013 to 2023. Publication output peaked in 2021 (41 papers), with minor dips in 2017, 2019, and 2022; annual citations peaked in 2022 (1,020) and decreased slightly in 2023 (970). Over the past five years, both indicators have remained at comparatively high levels. Notably, the increase in annual citations is steeper and has remained elevated over the past three years, which is consistent with citation accumulation and time-lag effects, suggesting that earlier contributions are gaining visibility and influence (Moed, 2005; Wang et al., 2013). The slight decline in 2023 likely reflects a shorter citation window and indexing latency rather than a genuine drop in scholarly activity.

Institutional landscape and international collaboration

Mapping inter-institutional co-authorship helps reveal the geographic patterns of collaboration and the organizational hubs that structure the field (Newman, 2001; Glänzel & Schubert, 2004; Chen, 2006). As shown in Fig. 3, the institutional collaboration network displays several regional clusters, with a few prominent hubs and many sparsely connected nodes, indicating a fragmented structure with limited cross-regional ties. The largest visible hub is Michigan State University, which is surrounded by multiple cross-institutional links. Other notable U.S. nodes include Pennsylvania State University (and the Pennsylvania Commonwealth System of Higher Education, if applicable), the University System of Georgia, and state university systems in the Western and Southeastern U.S. Outside the U.S., institutions such as Free Universität Berlin, Boğaziçi University, and Istanbul Aydın University appear as recognizable nodes, suggesting participation from Europe and Türkiye.

The top 10 institutions by publication count (Table 1) further substantiate the network-level pattern. Michigan State University ranks first (20 papers; earliest yea = 2013; betweenness centrality = 0.34), followed by University System of Georgia (13; 2015; 0.13) and the Pennsylvania Commonwealth System of Higher Education (PCSHE) (13; 2013; 0.03). Pennsylvania State University and its

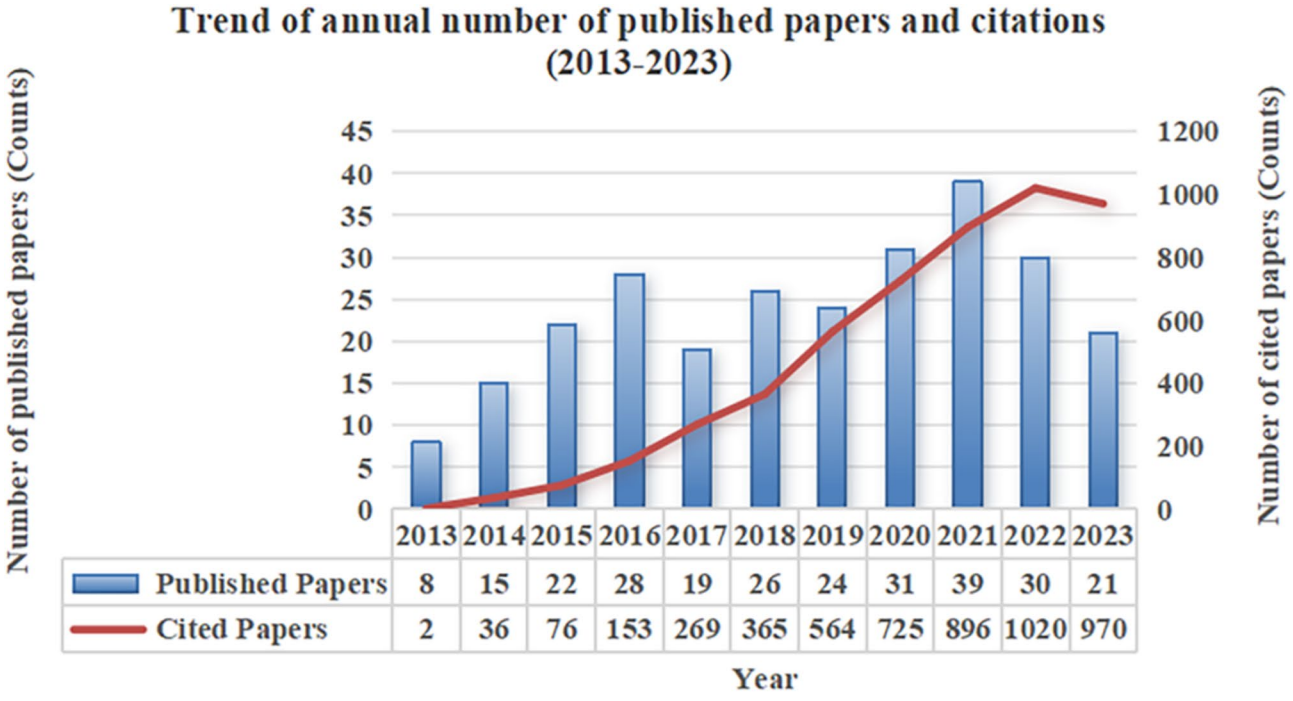


Fig. 2 Trends in annual publication output and citation counts in scientific practices research in science education (2013–2023, WoS core collection)

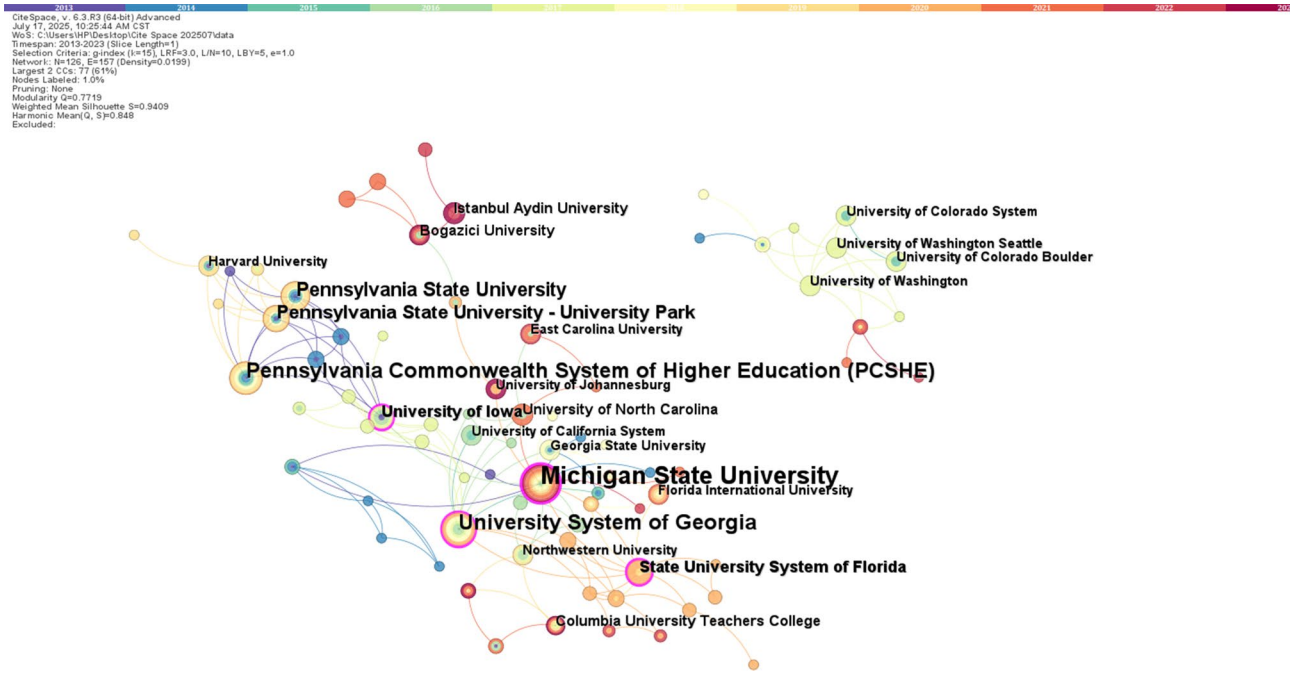


Fig. 3 International collaboration network of leading research institutions in scientific practices research in science education (2013–2023, WoS core collection)

University Park campus also rank among the leading contributors (10; 2013; 0.03 and 9; 2013; 0.03, respectively). Notably, the University of Iowa registers comparatively high betweenness centrality (6; 2013; 0.19), suggesting a bridging position in the collaboration network. Overall, these institutions anchor collaboration within regional clusters yet sustain only a modest number of cross-regional ties (Freeman, 1977; Brandes, 2001).

Overall, the institutional map indicates a concentration around a small number of U.S.-based hubs, with regional clustering elsewhere. Although the field includes international participation (e.g., Boğaziçi University

Table 1 Top 10 research institutions ranked by publication volume and their betweenness centrality in scientific practices research in science education (2013–2023, WoS core collection)

Number	Count	Institution	Year	Centrality
1	20	Michigan State University	2013	0.34
2	13	University System of Georgia	2015	0.13
3	13	Pennsylvania Commonwealth System of Higher Education (PCSHE)	2013	0.03
4	10	Pennsylvania State University	2013	0.03
5	9	Pennsylvania State University - University Park	2013	0.03
6	6	University of Iowa	2013	0.19
7	6	State University System of Florida	2019	0.12
8	5	University of North Carolina	2016	0.07
9	5	Bogazici University	2016	0.06
10	5	Columbia University Teachers College	2022	0.02

ranks among the top 10), the network's low connectivity and clustered configuration suggest substantial scope to strengthen cross-regional collaboration and to build broader inter-institutional coalitions (Wagner & Leydesdorff, 2005; Newman, 2004).

Keyword-based thematic mapping

Keyword co-occurrence clusters and top terms

To identify the thematic structure of the field in research on scientific practices, we conducted a clustering analysis of the keyword co-occurrence network derived from the sampled literature. The network comprises 161 keyword nodes and 677 co-occurrence links, with a network density of 0.0526. The clustering quality is satisfactory, as indicated by a modularity Q of 0.5665 and a mean silhouette score of 0.7963, suggesting a clear structure and reasonable cluster separation. Prior to generating the network and the high-frequency keyword list, we performed morphological normalization and synonym consolidation (e.g., harmonizing singular/plural forms and hyphenated variants, and removing stop words). Frequencies and betweenness centrality were computed based on the normalized keyword list. Cluster labeling used CiteSpace's latent semantic indexing (LSI), followed by manual review and semantic integration based on the relationships among high-frequency terms and the textual context of the sampled papers (Fig. 4a). The consolidation principles were (a) semantic proximity supported by strong inter-cluster links and (b) a preference for larger, more interpretable clusters. Accordingly, the initial eight clusters were merged into six thematic categories (T1–T6).

To make this two-stage structuring process transparent, Table 2 reports the original keyword co-occurrence clusters (#0–#7), their algorithm-generated cluster labels, representative high-frequency keywords, and our manual consolidation into six themes (T1–T6). Together with the cluster landscape (Fig. 4a), Table 2 provides the keyword evidence and consolidation rationale for the six themes discussed below.

To clarify the logical and structural relationships among these themes beyond composition, we further derived a theme-level coupling view from the keyword co-occurrence network exported as a GEXF file. Specifically, we aggregated node-level co-occurrence edges into inter-theme link counts (edge_count), defined as the number of co-occurrence links connecting keywords assigned to two different themes. This procedure provides a theme-level coupling profile and supports assessment of whether the field exhibits a core–support configuration. Figure 4b presents a core–support coupling map in a support–core layout, where edge thickness is proportional to edge_count and edge labels report the corresponding link counts. For readability, Fig. 4b displays links with edge_count ≥ 3. Because keyword co-occurrence ties are undirected, the layout is used for readability only and does not imply directionality or causal influence. The complete inter-theme edge list for all 15 theme pairs (including zeros) is reported in Table Sx (Supplementary Materials), enabling transparent inspection and replication of the structural interpretation; support-to-core linkage shares are reported in Fig. S1. Taken together, Fig. 4b and Table Sx indicate that cross-theme coupling concentrates around the core themes (T1–T3), whereas links among the support themes (T4–T6) are comparatively sparse (see also Fig. S1).

Co-occurrence ties are undirected, and the layout is for readability only.

To enhance the representativeness of our thematic interpretation and to ground the keyword-derived themes in the field's shared intellectual base, we triangulated the co-occurrence results with evidence from the document co-citation network. Figure 5 highlights an intellectual structure anchored in policy and framework texts (e.g., NRC/NGSS) and high-impact research on epistemic practices, argumentation, modeling, and assessment, while high-centrality core works are reported in Table S1 (Supplementary Materials). This triangulation helps calibrate our thematic narratives against dominant research traditions and reduces the risk of over-interpreting patterns driven primarily by smaller or context-specific keyword clusters.Against this triangulated backdrop, we interpret the consolidated keyword clusters as six themes (T1–T6) below.

T1 Scientific practices and classroom contexts (scientific practices, classroom) focuses on how scientific

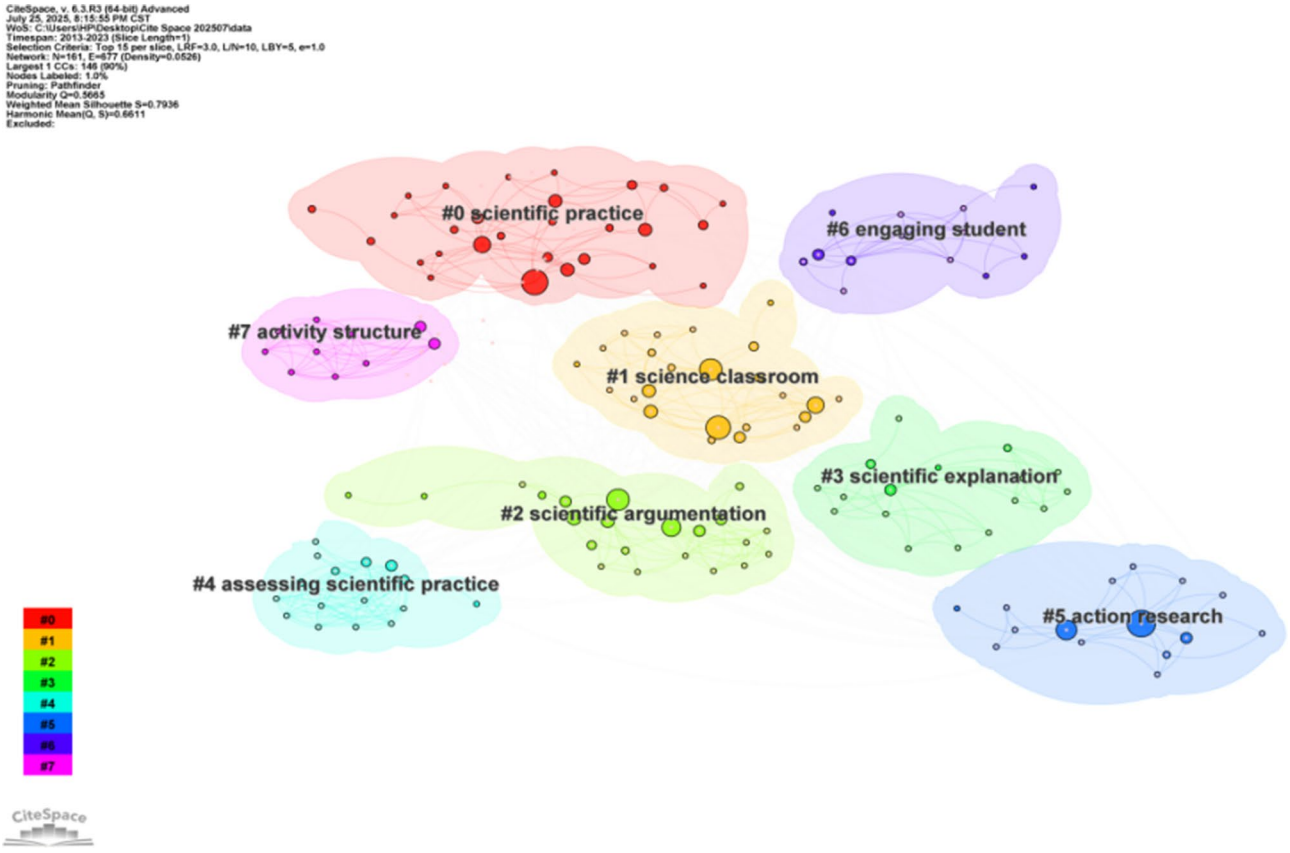


Fig. 4 (**a**) Keyword co-occurrence network and clustering map in scientific practices research in science education (2013–2023, WoS core collection). Nodes represent keywords and links indicate co-occurrence relations; cluster composition and consolidation into six themes (T1–T6) are reported in Table 2. (**b**) Core-support linkage map (edge thickness proportional to inter-theme link counts [edge_count]; links shown for edge_count ≥ 3)

practices are organized, negotiated, and sustained in classroom and laboratory settings. Its keyword set includes science classroom, inquiry, laboratory experience, teaching collaboration, and virtual laboratories (Table 2). This theme functions as a foundational and bridging domain for translating conceptual commitments about practices into implementable classroom and laboratory environments. First, research examines how teachers and students negotiate the meanings of practice and epistemic norms within particular instructional contexts (Stroupe, 2014; Miller et al., 2018); second, studies highlight the importance of making task purposes explicit and sustaining normative guidance through responsive instructional orchestration—conditions that appear especially critical for stable enactment in inquiry- or model-based laboratory instruction (Cooper et al., 2022); third, work on technology-mediated environments (e.g., virtual laboratories) suggests that digital settings can expand opportunities for students to participate in scientific practices during laboratory learning (Chan et al., 2021). In contrast to T6, which centers on task architectures as design mechanisms for enacting practice–content integration, T1 foregrounds the contextual conditions and interactional norms that enable and stabilize practice participation across classroom and laboratory settings. Overall, T1 offers a well-grounded account of in situ practice enactment; however, more systematic comparative research is needed to clarify the conditions that support transfer between classroom and laboratory settings.

T2 Scientific argumentation and scientific explanation (scientific argumentation, scientific explanation) examines how learners construct, evaluate, and revise claims based on evidence and reasoning, and how scientific meaning is generated through the processes of explanation and argumentation. Its keyword set includes argumentation, explanation, mechanistic reasoning, PCK, learning progression, and peer/group work (Table 2). This theme aims to strengthen the quality of epistemic performances and to identify teachable pathways in classroom settings. Research in this area can be summarized in three interrelated strands. First, the argumentation tradition exemplified by Osborne emphasizes norms for evidence and reasoning, as well as mechanistic reasoning, in improving students' argumentation and explanation (Osborne et al., 2004; Osborne, 2010). This line of work

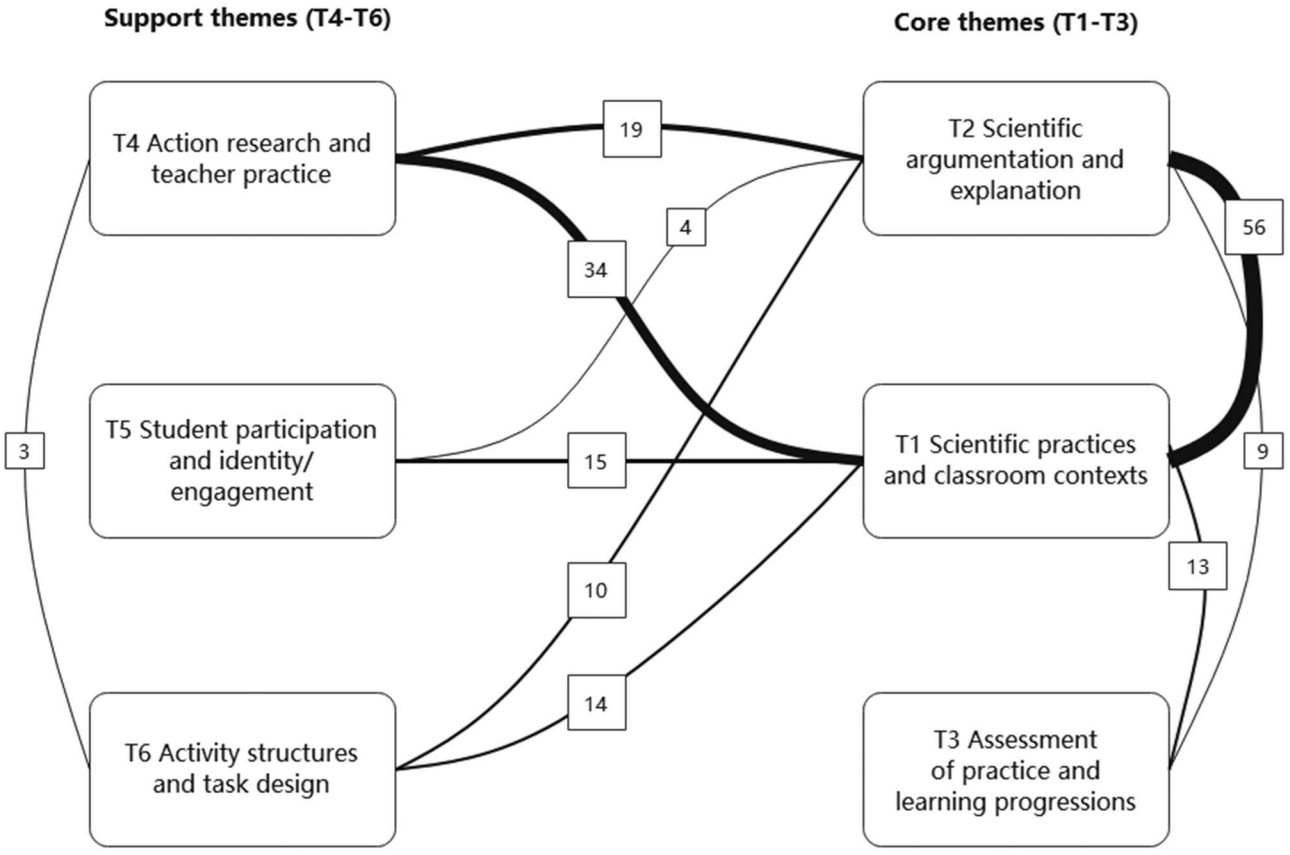


Fig. 4 (continued)

Table 2 Keyword co-occurrence clusters and consolidated themes in scientific practices education research (2013–2023)

Cluster ID	Size	Algorithm label (LSI)	Representative keywords (condensed)	Consolidated theme (manual)
#0	28	scientific practice	scientific practice; inquiry; general chemistry; laboratory experience; teaching collaboration; virtual laboratory; argument assessment	T1 Scientific practices and classroom contexts
#1	24	science classroom	science classroom; scientific modeling; model-based reasoning; science teaching; case study; heterogeneous nature	T1 Scientific practices and classroom contexts
#2	22	scientific argumentation	scientific argumentation; learning progression; framework; pedagogical content knowledge; mechanistic reasoning; peer/group work	T2 Scientific argumentation and explanation
#3	17	scientific explanation	scientific explanation; activity/task structure; elementary classroom; learning environment; sense-making; student engagement	T2 Scientific argumentation and explanation
#4	16	assessing scientific practice	assessing scientific practice; assessment; authentic science; science fair; student-driven projects; citizen science; machine-learning method	T3 Assessment of practice and learning progressions
#5	15	action research	action research; teachers; beliefs; professional development; modeling instruction; laboratory instruction; lab report	T4 Action research and teacher practice
#6	14	engaging student	engaging student; students' views; school science; science understanding; inquiry-based laboratories; undergraduate/graduate contexts	T5 Student participation and identity/engagement
#7	10	activity structure	activity structure; design; task design; problem-solving action; chemical education research; metacognitive tool	T6 Activity structures and task design

aligns with Berland and colleagues' analyses of how the goals and norms of classroom argumentation/explanation are constructed and negotiated in instructional contexts (Berland & Reiser, 2009; Berland & McNeill, 2010). Second, teachers' pedagogical content knowledge (PCK) for argumentation, together with dialogic scaffolding, is widely viewed as a key condition for high-quality student participation (McNeill & Knight, 2013; González-Howard & McNeill, 2019). Third, three-dimensional coherence-oriented redesign of tasks and assessments has been used to elicit more complete explanatory and argumentative performances and to improve their quality (Underwood et al., 2018). Meanwhile, studies also suggest that coordinating modeling with argumentation within peer/

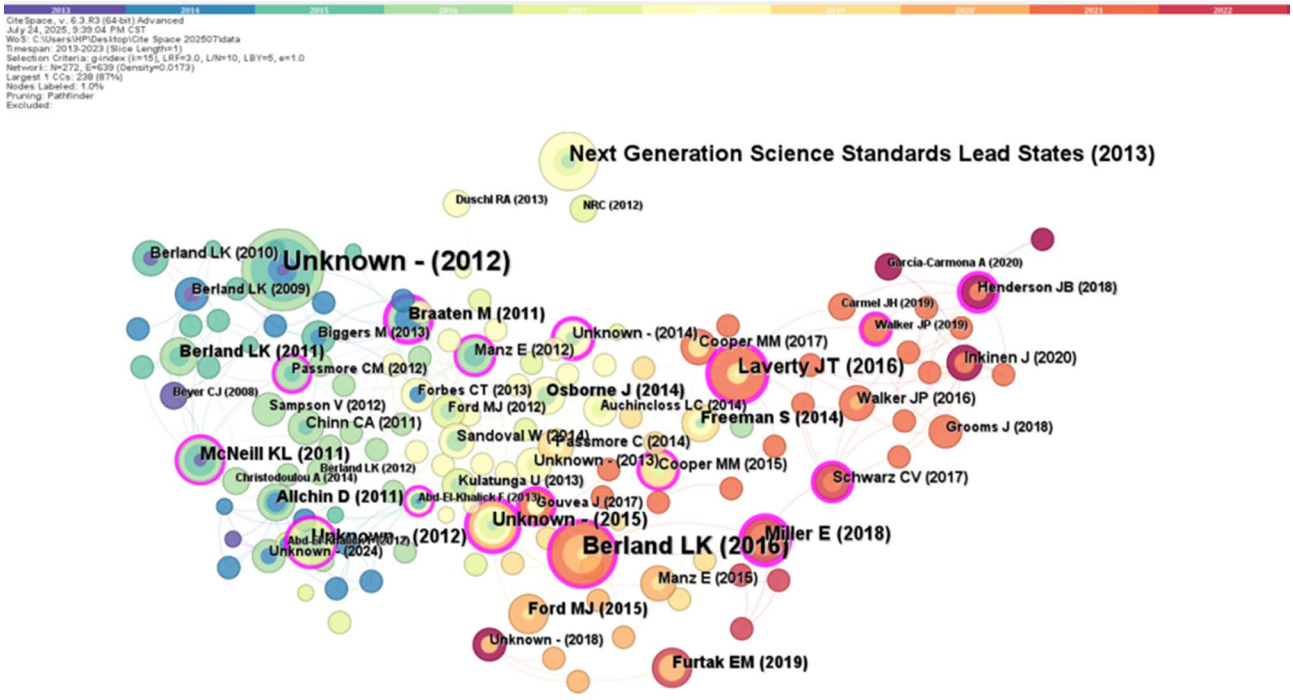


Fig. 5 Co-citation network of cited references in scientific practices research in science education (2013–2023, WoS core collection)

group interaction can yield reciprocal support between these practices and promote higher-quality outcomes (Evagorou et al., 2020). Overall, this theme has accumulated a relatively coherent evidence base; however, further integrative work is needed to clarify how teacher knowledge, task structures, and peer interaction jointly contribute to improvements in students' argumentation and explanation.

T3 Practice assessment and learning progressions (learning progression, framework, rubrics/validity, authentic science, citizen science, machine learning, trace/analytics) reflects a shift toward practice-oriented assessment within practice-oriented curricula. By developing structured assessment frameworks and building validity evidence, this line of work improves the usability and interpretability of assessment, while drawing on authentic tasks and technology-enhanced environments to broaden evidence sources (Table 2). This theme serves as a critical bridge for supporting coherence across curriculum, instruction, and assessment. Current research concentrates on three areas. First, the learning progression paradigm provides a reference model of "hierarchical hypotheses–iterative tool development–evidence calibration" for practice assessment, and many studies combine it with psychometric approaches (e.g., Rasch modeling) to strengthen measurement quality (Jin & Anderson, 2012; Cardace et al., 2021). Second, authentic contexts such as science fairs, student-driven projects, and citizen science are increasingly used to enhance ecological validity and to diversify the performances that assessments can capture (Koomen et al., 2018; Pizzolato & Tsuji, 2022). Third, approaches grounded in log/trace data and Evidence-Centered Design (ECD) support the construction of evidentiary chains that link process data to inferences about competence (Hao & Mislevy, 2018). Overall, T3 captures cutting-edge developments; however, a key challenge remains how to align and jointly calibrate evidence from authentic contexts and process-oriented data with learning progression models.

T4 Action research and teacher practice (teachers, beliefs, action research, professional development, modeling) focuses on teachers' instructional actions, belief systems, and the reflective practices they develop through action research during professional learning. It highlights the cyclical "plan–act–observe–reflect" process as a methodological foundation for evidence-informed instructional improvement (Kemmis & McTaggart, 1988; Cochran-Smith & Lytle, 2009) (Table 2). This theme foregrounds teachers' central role in the design and continuous refinement of practice-oriented instruction. Research under this theme converges on three interrelated emphases. First, studies show that teachers' epistemic understandings of scientific knowledge and practice shape task selection, classroom discourse organization, and assessment priorities. Second, integrating action research into teacher and preservice teacher education can strengthen the collection and use of evidence around classroom problems and support reflective improvement, thereby facilitating transfer into classroom teaching (Saribas & Akdemir, 2022; Al-Gaseem & Khasawneh, 2022). Third,

the development of instructional strategies for multiple practices–particularly modeling–often requires preservice teachers to engage in authentic practice and gradually build a more systematic awareness of scaffolding tools and support resources (Bennion & Davis, 2024). Overall, this line of research shows a coherent trajectory; however, further longitudinal evidence is needed to clarify how shifts in teachers' beliefs stabilize and translate into transferable design knowledge for practice-oriented instruction.

T5 Student engagement and identity and views (engaging students, students' views, school science, inquiry-based laboratories, undergraduate/graduate contexts) examines the quality of student participation, the formation of views, and the development of identity in school science and inquiry-based laboratory courses. It emphasizes that learning experiences should shift from merely "completing tasks" toward "building connections with the scientific community" (Table 2). This theme reflects a learner-centered perspective on practice participation. Current research can be characterized by three major orientations. First, studies investigate how students understand and revise their views about the value, authenticity, and epistemic meanings of school science and laboratory coursework (Sulaiman et al., 2023). Second, by making practice goals explicit and providing supportive learning structures, research shows that participation opportunities and academic outcomes can be improved for underserved groups (Killpack & Popolizio, 2023). Third, translating contemporary scientific practices into actionable classroom activity structures offers viable pathways for strengthening students' sense of connection to the scientific community and their identity-related experiences (Vamvakas et al., 2023). Overall, this theme has accumulated a solid empirical base; however, more systematic longitudinal research is needed to clarify how increased engagement develops into stable identities as participants in scientific practices and sustained academic or career trajectories.

T6 Activity structure and task design (activity structure, design, task design, problem-solving action, metacognitive tool) focuses on how practice–content integration is enacted through the design of activity sequences, task structures, and embedded cognitive/metacognitive tools. It emphasizes that coordinated designs of activity workflows and task-embedded scaffolds can enhance the usability of practice enactment and support transfer across instructional contexts (Table 2). This theme functions as a key mediating bridge that connects "practice ideals" with "classroom structures." Existing studies broadly converge on three lines of inquiry. First, classroom analyses suggest that variations in activity structures shape students' problem-solving pathways and their use of evidence, thereby influencing how practice enactment becomes visible in classroom work (Criswell & Rushton, 2014). Second, staged activity workflows and metacognitive scaffolds (e.g., Goldilocks Help) can promote mechanistic reasoning and self-regulatory capacities by making the phases and prompts of explanatory reasoning explicit (Yuriev et al., 2017). Third, learning-goal-driven design models and context-sensitive materials development work to transform authentic disciplinary practices (e.g., modeling) into teachable task sequences, while technology-enhanced analytic and formative-assessment approaches can capture process-level evidence of learners' reasoning from their task responses during enactment, thereby informing iterative refinement of task integration (Krajcik et al., 2008; Prins et al., 2018; Martin & Graulich, 2023). Overall, T6 has established a relatively coherent design–evidence linkage. However, further comparative research is needed to clarify the boundary conditions of task design principles across disciplines and grade levels, and to specify how designed scaffolds interact with practices such as modeling and argumentation in supporting learning outcomes.

Furthermore, several smaller clusters captured peripheral threads—such as context-specific, case-based work and cross-disciplinary or interdisciplinary settings. Given their sparse keyword coverage and limited connectivity in the co-occurrence network, these clusters were not included in the thematic consolidation.

Taken together, the cluster meanings (Table 2) and the inter-theme coupling pattern (Fig. 4b; Table Sx) support a core–support configuration of the field. T1–T3 form the core agenda, focusing respectively on classroom/laboratory enactment of practices (T1), argumentation and explanation as central epistemic performances (T2), and practice-oriented assessment and learning progressions (T3). In contrast, T4–T6 capture more proximal implementation conditions and mechanisms–teacher practice and professional learning (T4), student engagement/identity and participation conditions (T5), and activity/task structures and design tools (T6). Importantly, the strongest cross-theme connections are concentrated around the core set, and the cross-theme ties of T4–T6 connect predominantly to T1–T3 (Fig. 4b; Table Sx), indicating that these micro-level themes are structurally oriented toward enabling, mediating, and scaling the core research agenda in instructional contexts. This pattern is also reflected in the support-to-core linkage shares computed from inter-theme link counts (T4: 90.3%, T5: 82.6%, T6: 86.7%; Fig. S1).

Building on this thematic framework, we further examined high-frequency terms and betweenness centrality (BC) to corroborate the six themes identified through clustering. As shown in Table 3, inquiry and science education appear most frequently (each 76 occurrences), followed by science (69), knowledge (48), and students (45),

Table 3 Top 20 high-frequency keywords and their betweenness centrality in scientific practices research in science education (2013–2023, WoS core collection)

Ranking	Frequency	Centrality	Year	Keyword	Rankin	Frequency	Centrality	Year	Keyword
1	76	0.10	2013	inquiry	11	17	0.02	2014	teachers
2	76	0.11	2013	science education	12	16	0.11	2013	explanation
3	69	0.14	2013	science	13	14	0.11	2013	views
4	48	0.14	2013	knowledge	14	13	0.03	2013	classroom
5	45	0.12	2013	students	15	12	0.08	2013	school science
6	44	0.03	2013	scientific practices	16	12	0.01	2018	chemical education research
7	39	0.19	2013	instruction	17	11	0.05	2013	learning progression
8	35	0.12	2013	argumentation	18	11	0.20	2013	beliefs
9	22	0.06	2013	framework	19	9	0.02	2013	design
10	20	0.20	2013	model	20	9	0.12	2013	conceptual change

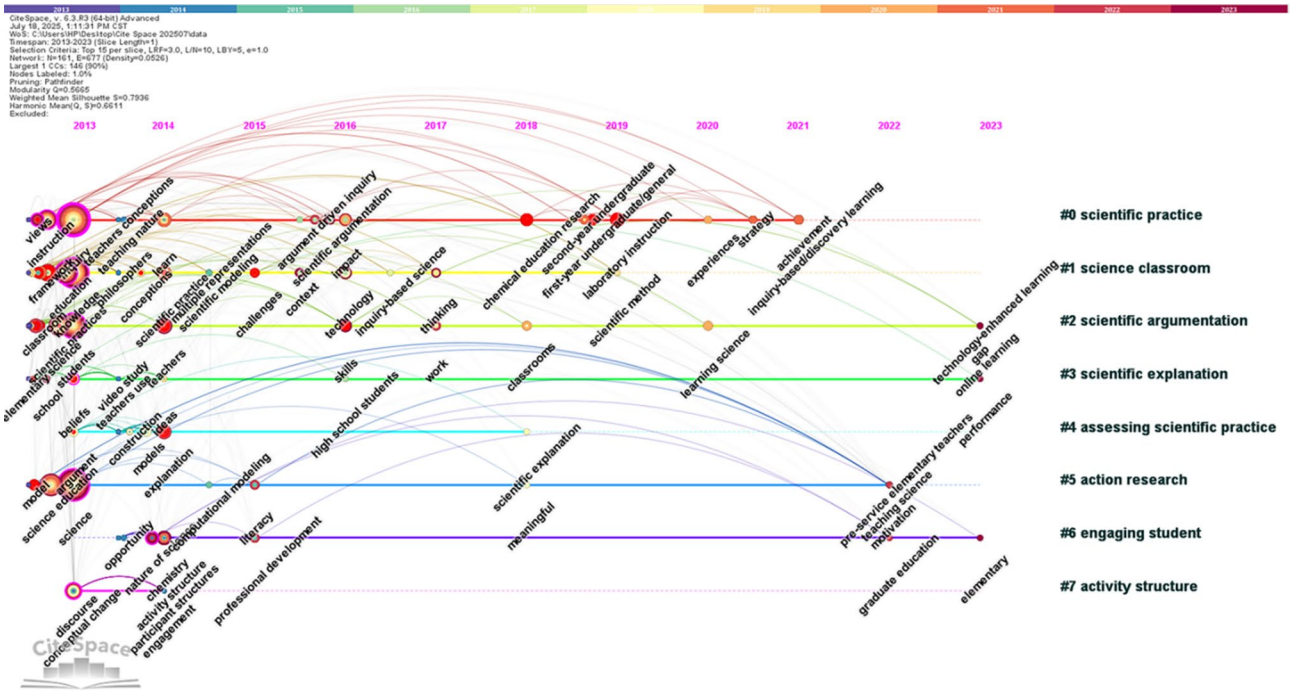


Fig. 6 (**a**) Keyword timeline visualization in scientific practices research in science education (2013–2023, WoS core collection). (**b**) Keyword burst detection results (top 23) in scientific practices research in science education (2013–2023, WoS core collection)

which together constitute the semantic core of the field. In terms of BC, model (0.20), instruction (0.19), knowledge (0.14), argumentation (0.12), and explanation (0.11) exceed a commonly used salience threshold (≥0.10), indicating their bridging roles across clusters. Taken together, Table 2 provides the cluster-based thematic evidence, whereas Table 3 offers a complementary frequency-and-centrality perspective that corroborates the field's semantic core and cross-theme connectors. Consistent with the inter-cluster connections shown in Fig. 4a, these results suggest that classroom practice provides a basis for advancing evidence-based reasoning and knowledge construction. In parallel, assessment–progression mechanisms create a feedback loop for improvement, while teacher practice, student engagement, and task design provide implementation scaffolds that characterize the field's dominant research trajectory.

Thematic timelines

As shown in Fig. 6a, the timeline view of the keyword co-occurrence map indicates that research on scientific practices from 2013 to 2023 unfolded in four successive, partially overlapping stages. In this visualization, clusters are arranged horizontally. Each node denotes a keyword: its horizontal position marks the time slice in which it first appears, while node size reflects its overall frequency in the corpus. Links indicate keyword co-occurrence relations. To enhance interpretive transparency and reproducibility, we also report keyword burst-detection results (Fig. 6b), which provide burst intervals (Begin–End) and

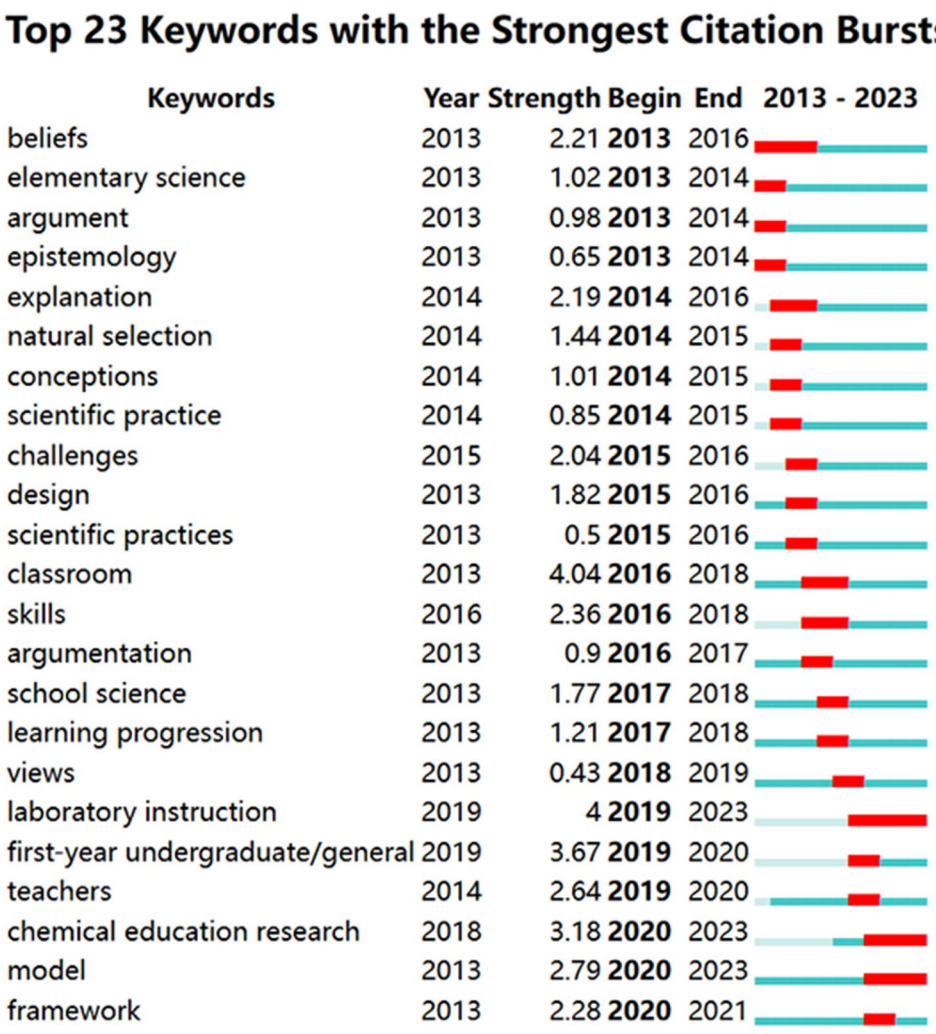


Fig. 6 (continued)

strength values for terms exhibiting abrupt increases in attention. Accordingly, Fig. 6a supports a "when-appears/where-clusters/how-connects" reading, whereas Fig. 6b is referenced when describing term-level surges.

(1) 2013–2015: Foundational phase. Research during this period primarily focused on articulating pedagogical aims and proposing early curricular framings for scientific practices. In Fig. 6a, high-frequency nodes are dominated by foundational and classroom-context terms (e.g., beliefs, elementary science), which first appear and remain salient in the early slices (≈2013–2015). Consistent with this pattern, burst detection in Fig. 6b identifies beliefs (2013–2016; Strength = 2.21) and elementary science (2013–2014; Strength = 1.02) as early bursts. Overall, the keyword profile suggests sustained attention to baseline principles, instructional goals, and classroom settings for enacting practices, aligning most closely with T1/T6.

(2) 2015–2019: Reasoning-deepening phase. During this period, epistemic-practice terms such as argumentation and explanation became prominent, and argument-driven inquiry emerged as a salient mid-2010s term (Fig. 6a). Burst detection offers corroborating evidence of increased attention to reasoning-related constructs, including explanation (2014–2016; Strength = 2.19), which bridges the late foundational and early reasoning-deepening slices, and argumentation (2016–2017; Strength = 0.90) (Fig. 6b). The keyword profile suggests a shift toward designing inquiry processes and supporting students' claim–evidence–reasoning (CER) work, aligning most closely with T2. At the same time, cross-cutting terms—most notably model and instruction—recur across slices and appear in multiple clusters, linking classroom enactment with epistemic practices (Fig. 6a). Looking ahead to the most recent slices, model exhibits a renewed burst (2020–2023; Strength = 2.79; Fig. 6b), consistent

with its cross-cluster bridging role and with the betweenness-centrality patterns reported in Table 3.

(3) 2018–2020: Assessment and improvement phase. In the late 2010s, assessment- and improvement-oriented strands became more visible in the timeline view (Fig. 6a). Burst detection provides a transitional signal in this direction: learning progression shows a brief burst (2017–2018; Strength = 1.21), bridging into the subsequent 2019–2020 increase in population- and context-specific terms (Fig. 6b). As the timeline enters the 2019–2023 window, attention concentrates on practice-related instructional settings, most notably laboratory instruction (2019–2023; Strength = 4.00), alongside terms that foreground focal populations and implementation contexts such as first-year undergraduate/general (2019–2020; Strength = 3.67) and teachers (2019–2020; Strength = 2.64) (Fig. 6b). Together with the dedicated clusters labeled assessing scientific practice and action research in the timeline map (Fig. 6a), these patterns suggest a growing focus on making practices more accessible and improvable through instructionally actionable designs, aligning most closely with T3/T4.

(4) 2020–2023: Technology integration and disciplinary contextualization phase. In the rightmost time slices, technology- and context-infused keywords (e.g., technology-enhanced learning, online learning) became more visible in the timeline view (Fig. 6a), indicating heightened attention to digitally mediated enactment and hybrid learning settings in the most recent period captured by our corpus. Burst detection further shows that this late-stage signal is accompanied by stronger disciplinary anchoring and organizing constructs, including chemical education research (2020–2023; Strength = 3.18), model (2020–2023; Strength = 2.79), and framework (2020–2021; Strength = 2.28) (Fig. 6b). Taken together, these patterns–appearing alongside continued activity in argumentation/explanation- and assessment-related clusters in the timeline map (Fig. 6a)–suggest that practice enactment is increasingly situated in discipline-specific designs and organizing frameworks and, in some contexts, explored through digital infrastructures. This late-stage profile aligns most closely with T5/T6, while also intersecting with T2/T3.

In summary, the combined evidence from the timeline view and burst detection indicates shifting emphases across the 2013–2023 corpus: from early attention to theoretical and curricular framing, to heightened salience of evidence-based reasoning and inquiry enactment, followed by a stronger focus on assessment-oriented refinement and instructional contextualization, and–toward the latest slices–to technology-mediated enactment alongside organizing constructs. This temporal interpretation is consistent with the T1–T6 thematic structure reported in Sect.Keyword co-occurrence clusters and top terms Corpus Characteristics Based on Manual Coding. In particular, the later period (especially 2019–2023) shows heightened attention to laboratory instruction, population- and context-specific settings (e.g., first-year undergraduate/general and teachers), and discipline-anchored constructs such as model and frameworks, alongside the visibility of digitally mediated contexts (Fig.s. 6a–6b). Notably, these stages overlap rather than replace one another, suggesting cumulative layering of research attention rather than discrete transitions.

Content-analytic structure of the research

Corpus characteristics based on manual coding

Study participants by educational level As shown in Fig. 7, the corpus is concentrated in school-based and undergraduate settings. Upper secondary (high school) studies account for 20.5% and undergraduate studies for 29.3% (49.8% combined). Within compulsory education, primary contributes 14.1% and lower secondary (middle school) 9.9%, whereas preschool/early childhood is rarely represented (1.5%). Teacher-focused studies are less common: work targeting in-service teachers constitutes 12.2%, and preservice- teacher studies 3.8%. Cross-level designs are uncommon (K–12: 3.4%; K–16: 0.8%), and 4.6% of studies fall into an other/unspecified category. Overall, student-focused, school-based research (including K–12/K–16 cross-level designs) accounts for roughly half of the sample, whereas teacher-focused work represents about one-sixth.

Distribution of disciplines As shown in Fig. 8, integrated/general science constitutes the largest share of the corpus (42.97%). Among the core disciplines, biology accounts for 21.29%, chemistry 18.25%, and physics 10.27% (≈49.8% combined). Geography is less frequently represented (3.04%), and the "other" category, dominated by environmental science, engineering, and technology, totals 4.18%. Overall, the distribution indicates that the literature concentrates on integrated science and core disciplines, with comparatively limited attention to engineering/technology and broader cross-disciplinary themes.

Typology and distribution of research questions

To analyze the structure of the research questions in scientific-practices studies, we coded 263 sampled articles using a content classification framework (codebook) developed for this review (Appendix S2-A). The coding

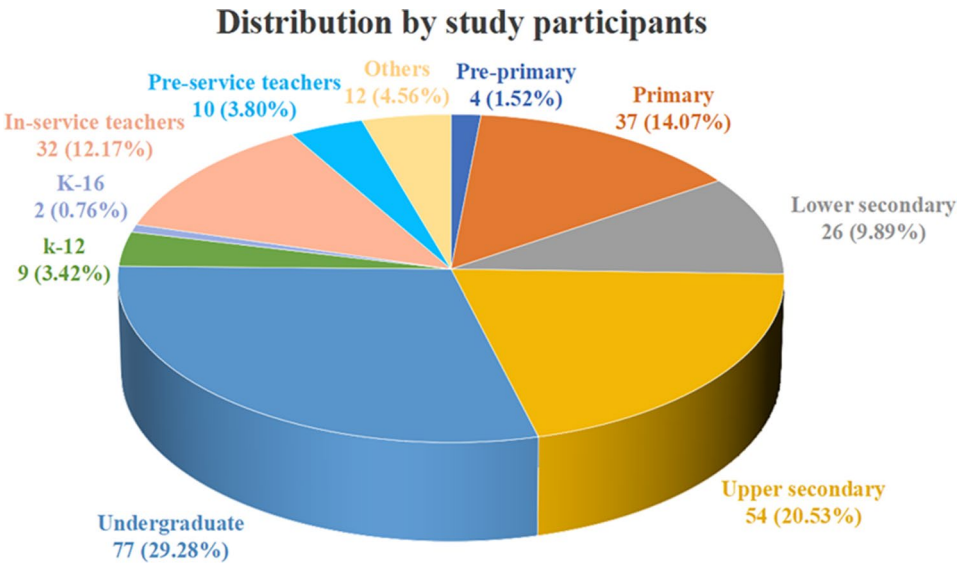


Fig. 7 Distribution of the educational stages of research participants in scientific practice research in science education (2013–2023, WoS core collection)

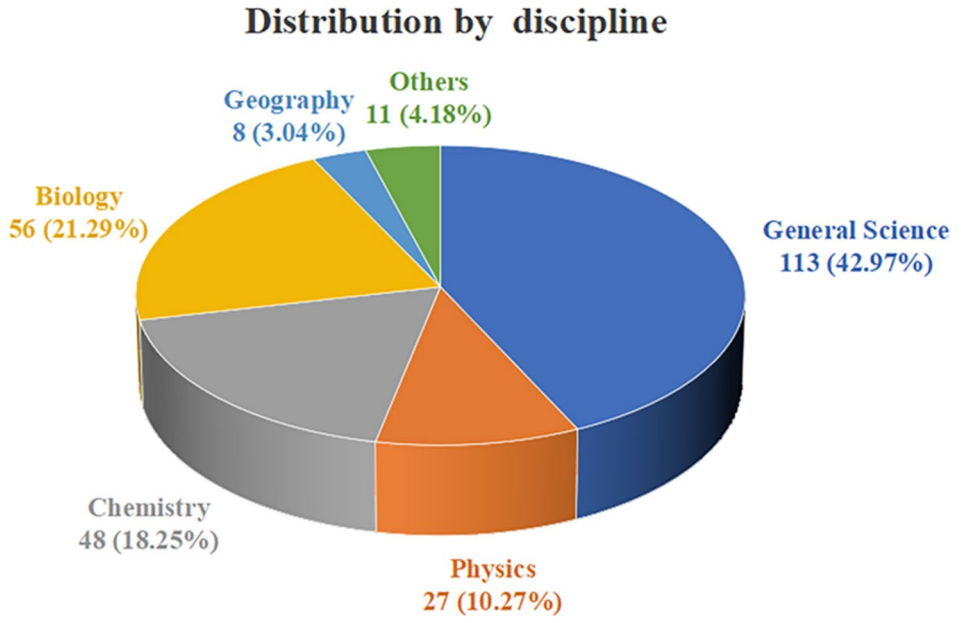


Fig. 8 Disciplinary distribution in scientific practice research in science education (2013–2023, WoS core collection)

unit was the explicitly stated research question. When an article did not state a research question, or stated it only implicitly, we inferred the focal question from the title and abstract. To ensure cross-study comparability, we used a single-label primary assignment with a preregistered priority rule. Specifically, if a question contained highly diagnostic terms (e.g., teacher development, teacher knowledge, identity, agency, equity) and the study aim was not primarily about assessment/measurement or instructional design/implementation, we assigned it to the corresponding conceptual category. When a question plausibly fit multiple categories, we selected the primary category according to the preset priority order (D > E >B >C >A >F >G). Category definitions and illustrative examples are provided in Table 4.

Using this framework, we classified research questions into seven categories: (A) effectiveness and comparison; (B) design and implementation; (C) processes and mechanisms; (D) assessment and measurement; (E) learning progressions and development; (F) teacher development/teacher knowledge; and (G) identity, agency, equity.

Before describing the results, we note that the following statistics provide a cross-sectional summary of the sampled corpus and are not intended to imply longitudinal

Table 4 Classification system for research questions in scientific practice research in science education and their typical examples

Category	Description (per Appendix)	Typical research question examples
A. Effectiveness/Comparison	Whether an intervention or context produces differences or improvement; comparisons with alternative approaches	In an online-simulator photosynthesis lab, can students effectively design experiments and collect data? Are outcomes from the virtual lab comparable to those from a physical lab?[15] (Sherrer, 2020).
B. Design/Implementation	How courses/units are designed, enacted, or integrated; task design and instructional strategies	To what extent does embedding scientific practices (via the BRH model) and model–evidence link (MEL) discussions into science/technology teaching courses influence preservice or in-service teachers' understanding of scientific methods and the use of evidence in practice?[16] (Saribas, 2022).
C. Process/Mechanism	Students' construction, reasoning, interactional processes, and underlying mechanisms, including discourse and representations	In modeling-based versus traditional elementary science lessons, what epistemological messages are conveyed through teacher talk?[17] (Oh et al., 2022).
D. Assessment/Measurement	Development of evaluation frameworks, rubrics, or instruments and their reliability/validity	How can an assessment tool be designed to cover the full cycle of investigation design → explanation → argumentation while aligning tightly with the learning objectives of high school Chemistry I?[18] (Hosbein et al., 2021).
E. Learning progression/Development	Developmental trajectories or progressions of students' knowledge or disciplinary practices	How does students' understanding of the timescale required for biological evolution progress from "individual life span" to "accumulation across generations"?[19] (Wyner et al., 2020).
F. Teacher development/Teacher Knowledge	Teachers' (pre-/in-service) knowledge, practice, and professional growth	During the implementation of practice-oriented science units, what challenges do teachers encounter, and how do mentoring and peer support help them overcome these and develop new instructional strategies?[20] (Benedict-Chambers et al., 2017).
G. Identity/Agency/Equity	Learners' identity construction, agency, sense of belonging, and issues of equity/justice	In inquiry-based science courses, how do students from diverse backgrounds experience participation, exercise agency, and develop a sense of belonging? What classroom structures support equitable participation and outcomes?[21] (Killpack & Popolizio, 2023).

trends. The distribution of research-question categories is shown in Fig. 9. In this cross-sectional snapshot of the sampled corpus, assessment and measurement (D) is the most prevalent category, accounting for 57.0% of items (150/263). The next most common category is design and implementation (B) at 25.5% (67/263), reflecting the field's strong concentration–within this dataset–on instructional design/implementation and closely related evaluative work (e.g., outcome validation, instrument development, and task construction). By contrast, learning progressions and development (E) and teacher development/teacher knowledge (F) appear less frequently, representing 8.7% (23/263) and 7.2% (19/263), respectively. Three categories are rare in the sample: processes and mechanisms (C) (0.8%, 2/263), effectiveness and comparison (A) (0.4%, 1/263), and identity/agency/equity (G) (0.4%, 1/263). Taken together, the distribution suggests that mechanistic accounts of learning and sociocultural dimensions are relatively underrepresented in this corpus. These low-frequency categories should be interpreted cautiously given the very small cell sizes.

Typology and distribution of methods

To identify the methodological orientations of the sampled literature, we used a triangulated classification procedure. For each article, we cross-checked the Methods section, abstract, and title keywords, and assigned a single primary method label using a preregistered priority rule to determine the label (M4 > M6 > M7 > M5 > M3 > M1 > M2 > M8). We distinguished eight method categories (Appendix S2-B): M1 quantitative empirical research; M2 qualitative research; M3 mixed-methods research; M4 design-based or action research; M5 instrument/assessment/technology development; M6 reviews (bibliometric/systematic) or meta-analytic studies; M7 theoretical/conceptual or framework papers; and M8 other/unspecified.

Based on the 263 sampled articles (Fig. 10; Table 5), design-based/action research (M4) is the most prevalent category, accounting for 31.2% (82/263). In this corpus, M4 studies most often take the form of practice-oriented classroom interventions and curricular innovation. Theoretical/conceptual or framework papers (M7) constitute 22.8% (60/263), reflecting substantial attention to model building and conceptual clarification. Instrument/assessment/technology development (M5) represents 15.6% (41/263), highlighting assessment tools, task design, and platform development as common pathways for translating practice-oriented ideas into usable resources. Quantitative empirical studies (M1) account for 14.1% (37/263), representing a substantive–though not dominant–share of data-driven evidence in the sample. Reviews/bibliometric or meta-analytic studies (M6) make up 9.9%

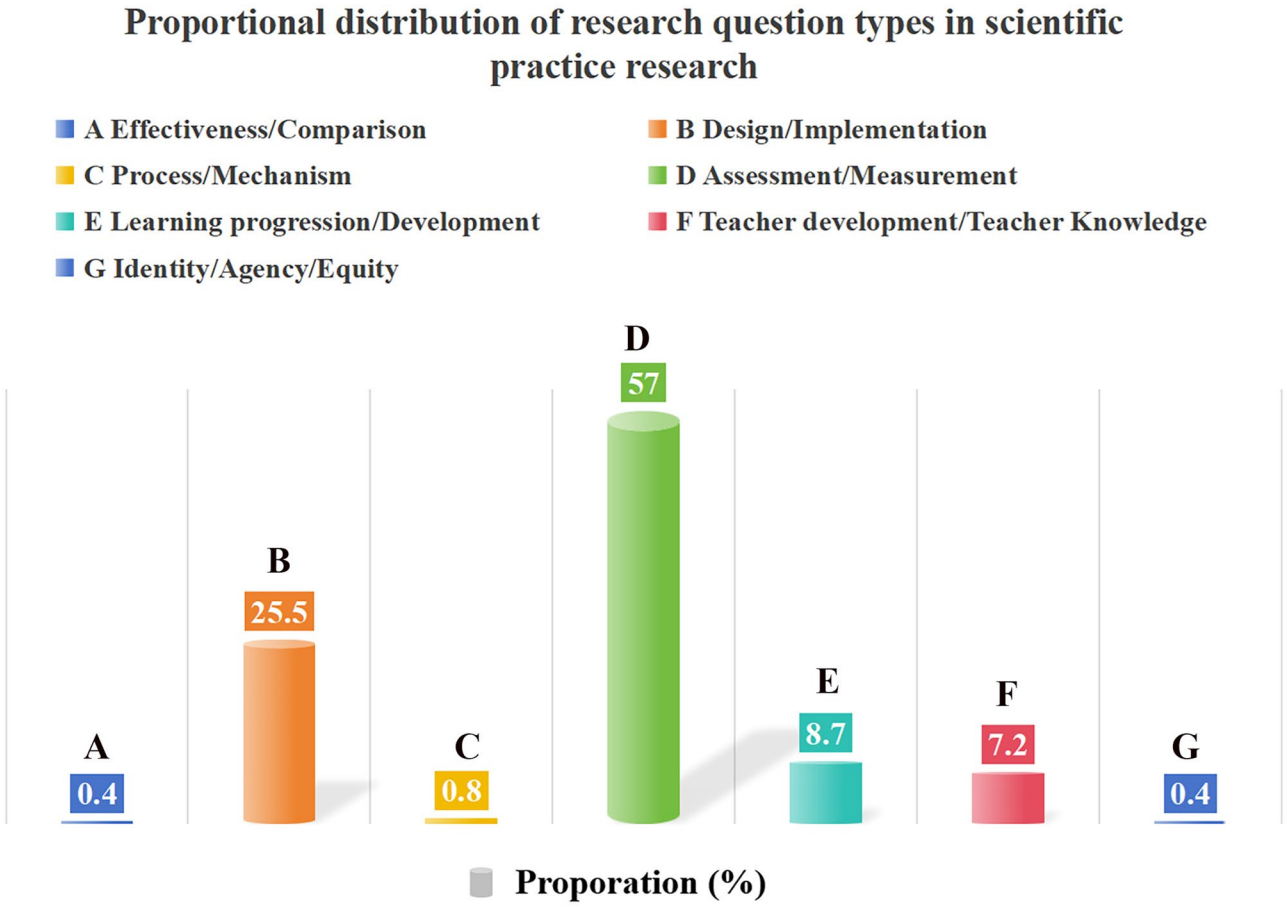


Fig. 9 Proportional distribution of research question types in scientific practice research in science education (2013–2023, WoS core collection)

(26/263), forming a visible component of synthesis- and mapping-oriented work within this dataset.

By contrast, qualitative studies (M2) (3.4%, 9/263) and mixed-methods studies (M3) (2.3%, 6/263) constitute relatively small shares of the sample. Within this corpus, the limited presence of qualitative designs suggests fewer studies that foreground fine-grained accounts of learning processes, classroom interaction, and contextual complexity. Likewise, the small proportion of mixed-methods work suggests that integrative designs combining multiple data sources and analytic lenses are uncommon in the dataset. These observations should be interpreted cautiously given the small cell sizes for M2 and M3.

A secondary label analysis further shows that method choices are concentrated within several subtypes. Across the sampled corpus, framework/model building (Theor1), interview/observation/classroom-interaction analysis (Qual1), and grounded/thematic content analysis (Qual3) are commonly used across multiple primary method categories. By contrast, questionnaire/statistical modeling (Q2) and experimental/quasi-experimental designs (Q1) appear mainly in quantitative empirical studies (M1) and instrument/assessment development studies (M5) (Table 5). Overall, this dataset reflects methodological diversity, and remaining centered on practice-oriented designs, supported by substantial theoretical work and quantitative analytic approaches.

Typology and distribution of theoretical/analytical frameworks

To identify the theoretical and analytical frameworks used in the sampled literature, we used a triangulated coding procedure. For each paper, we cross-checked the theory/framework section, abstract, keywords, and assigned a single primary framework label using a pre-registered priority rule (Appendix S2-C). The resulting taxonomy comprises six categories (Fig. 11; Table 6): F1 theoretical and conceptual frameworks; F2 inquiry- and design-based research frameworks; F3 assessment and measurement frameworks; F4 sociocultural and communication frameworks; F5 technology/tool-mediated frameworks; and F6 other/descriptive frameworks.

Based on 263 sampled articles (Fig. 11; Table 6), inquiry/design frameworks (F2) constitute the largest share (30.8%, 81/263). In this corpus, F2 studies typically combine inquiry-based learning (IBL) with design-based

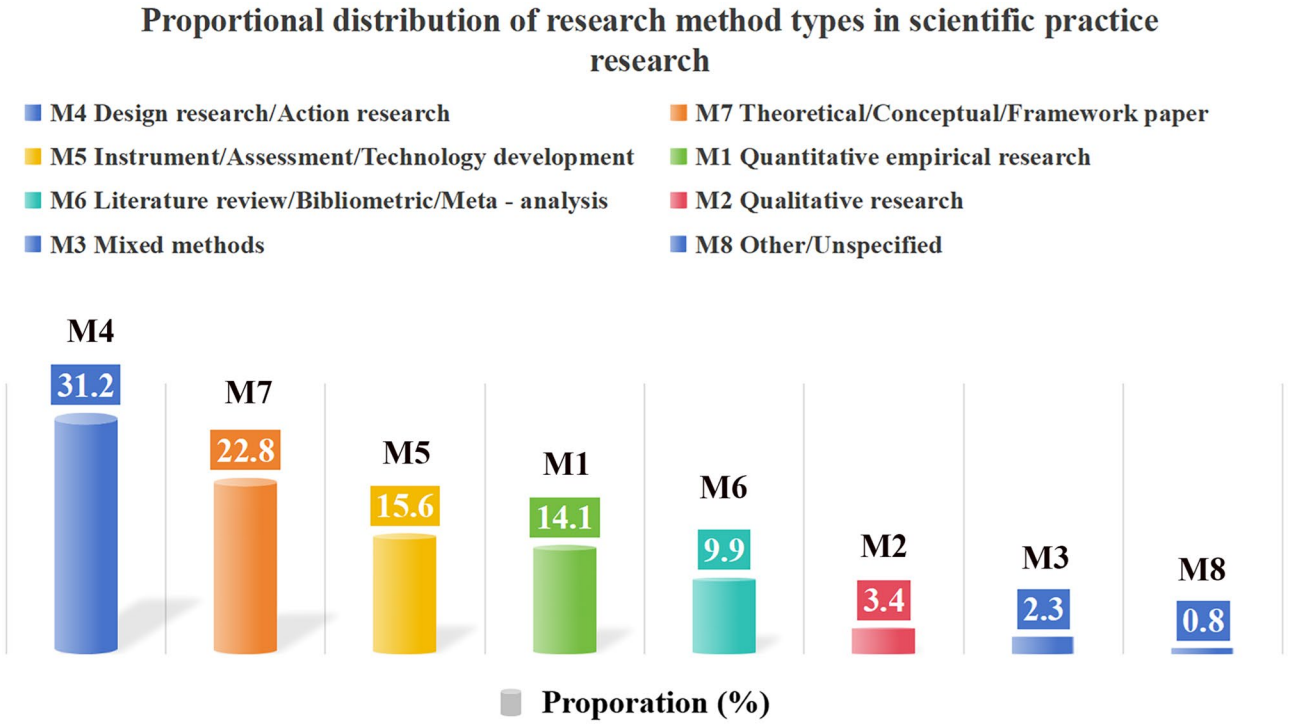


Fig. 10 Proportional distribution of research method types in scientific practice research in science education (2013–2023, WoS core collection)

Table 5 Classification structure of research methods and top 3 submethod types in scientific practice research in science education

Primary method (L1)	Count	Percentage (%)	Top 3 commonly used submethods (count; coverage %)
M4 Design research/Action research	82	31.2	Theor1 framework/model building (69; 84.1%); Qual1 interview/observation/classroom-interaction analysis (59; 72.0%); Qual3 grounded/thematic content analysis (51; 62.2%)
M7 Theoretical/Conceptual/Framework paper	60	22.8	Theor1 framework/model building (53; 88.3%); Qual3 grounded/thematic content analysis (38; 63.3%); Qual1 interview/observation/classroom-interaction analysis (38; 63.3%)
M5 Instrument/Assessment/Technology development	41	15.6	Q1 experimental/quasi-experimental design (32; 78.0%); Theor1 framework/model building (32; 78.0%); Q2 questionnaire/statistical modeling (30; 73.2%)
M1 Quantitative empirical research	37	14.1	Q2 questionnaire/statistical modeling (27; 73.0%); Q1 experimental/quasi-experimental design (25; 67.6%); Theor1 framework/model building (24; 64.9%)
M6 Literature review/Bibliometric/Meta-analysis	26	9.9	Theor1 framework/model building (21; 80.8%); Q1 experimental/quasi-experimental design (15; 57.7%); Q2 questionnaire/statistical modeling (15; 57.7%)
M2 Qualitative research	9	3.4	Qual1 interview/observation/classroom-interaction analysis (8; 88.9%); Theor1 framework/model building (8; 88.9%); Qual3 grounded/thematic content analysis (7; 77.8%)
M3 Mixed methods	6	2.3	Qual1 interview/observation/classroom-interaction analysis (5; 83.3%); Qual3 grounded/thematic content analysis (5; 83.3%); Q1 experimental/quasi-experimental design (4; 66.7%)
M8 Other/Unspecified	2	0.8	—

Note. Coverage is the proportion of studies within each L1 category that used the submethod (studies may use multiple submethods; thus, percentages can exceed 100%). Code key: Theor1 = framework/model building; Qual1 = interview/observation/classroom-interaction analysis; Qual3 = grounded/thematic content analysis; Q1 = experimental/quasiexperimental design; Q2 = questionnaire/statistical modeling

research or implementation research (DBR/DBIR) to foreground contextualized tasks and students' participation in practices. Sociocultural/communication frameworks (F4) rank second (23.6%, 62/263) and are commonly used to examine identity work, discourse/interaction, and equity, drawing on perspectives related to discourse, identity/agency, and the social nature of scientific practice. Technology/tool frameworks (F5)

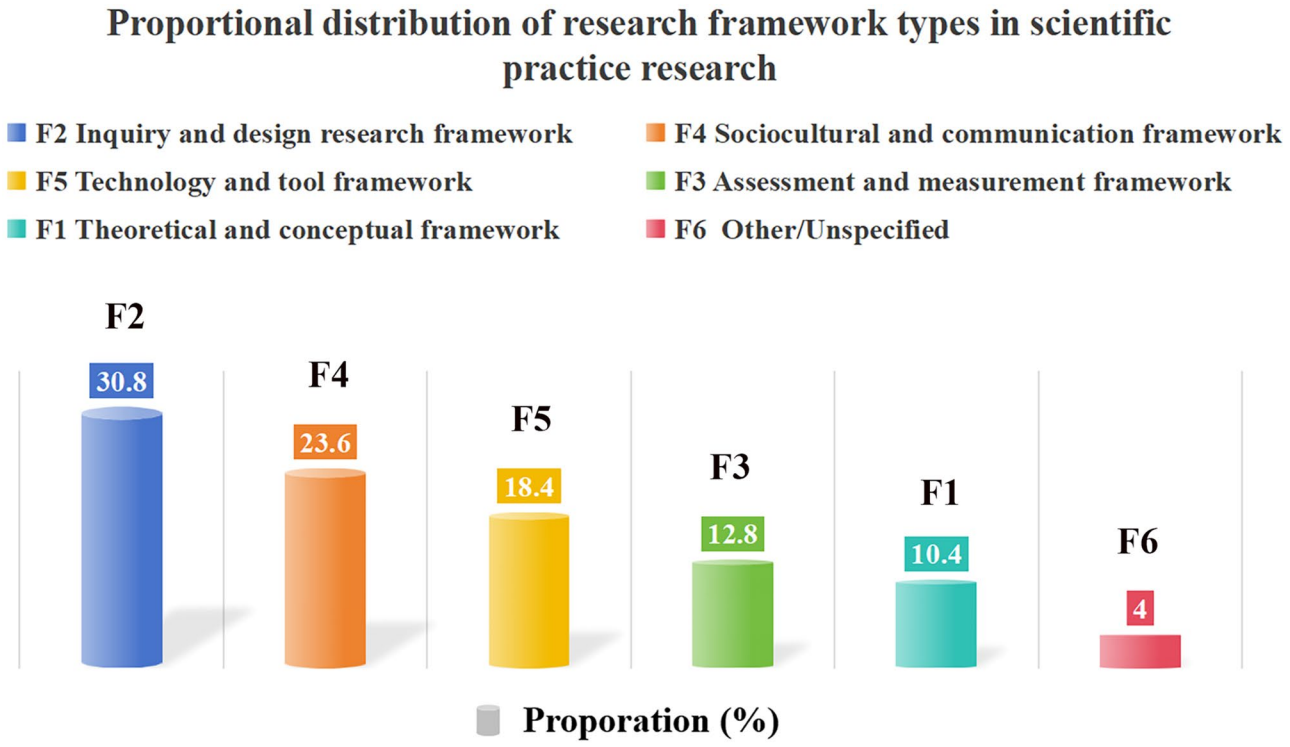


Fig. 11 Proportional distribution of theoretical and analytical framework types in scientific practice research in science education (2013–2023, WoS core collection)

Table 6 Classification of research frameworks and distribution of the top 3 subframeworks in scientific practice research in science education

Primary framework (L1)	Count	Percentage (%)	Top 3 commonly used subframeworks (count; coverage % within L1)
F2 Inquiry and design research framework	81	30.8	IBL inquiry-based learning (33; 40.7%); modeling/computational modeling (26; 32.1%); DBR/DBIR design-/implementation-based research (14; 17.3%)
F4 Sociocultural and communication framework	62	23.6	Discourse/interaction & discourse adaptation (18; 29.0%); identity/agency/equity (17; 27.4%); public engagement/Science communication (13; 21.0%)
F5 Technology and tool framework	48	18.3	Platforms/Systems/Instruments (32; 66.7%); learning analytics/adaptive/AI-assisted systems (12; 25.0%); virtual/online labs & simulation (5; 10.4%)
F3 Assessment and measurement framework	34	12.9	Validity/reliability evaluation frameworks (17; 50.0%); CER (claim–evidence–reasoning) & rubric schemes (7; 20.6%); 3D-LAP/three-dimensional learning assessment (6; 17.6%)
F1 Theoretical and conceptual framework	27	10.3	General learning theories & modeling accounts (17; 63.0%); epistemology/knowledge integration (9; 33.3%); three-dimensional learning/SEP framework (6; 22.2%)
F6 Other/Unspecified	11	4.2	—

Note. "Coverage" denotes the proportion of papers within each framework category that adopt the subframework. Because a single study may use multiple subframeworks, coverage percentages can exceed 100%

account for 18.3% (48/263) and are used to analyze how digital resources (e.g., virtual labs, AI-assisted systems, visualization platforms) mediate the teaching and learning of scientific practices.

By comparison, assessment/measurement frameworks (F3) and theoretical/conceptual frameworks (F1) appear less frequently in the sample, accounting for 12.9% (34/263) and 10.3% (27/263), respectively. In this corpus, F3 studies commonly draw on validity and reliability theory, claim-evidence-reasoning (CER), and rubric-based schemes to support coherence and operationalized capability assessment. F1 studies, in turn, foreground theoretical accounts of scientific learning and epistemology, as well as three-dimensional learning models (e.g., the Science and Engineering Practices in the Next Generation Science Standards, SEPs). In addition, 11 papers (4.2%) do not explicitly specify a framework or

rely primarily on descriptive framing and were classified as F6.

Overall, the distribution in this sampled corpus shows relatively higher shares of instructional design–related (F2), sociocultural/communication (F4), and technology/tool-mediation (F5) frameworks. Assessment/measurement frameworks (F3) account for 12.9% of framework labels. In this dataset, this modest share suggests that explicit linkages among assessment instruments, formal assessment theory, and interpretive models are not consistently articulated.

Multidimensional associations among research questions, methods, and frameworks (ASRs)

To examine structural coupling among research questions (RQ), methods (M), and theoretical/analytical frameworks (F), we constructed three contingency tables (RQ × M, RQ × F, and M × F) and computed adjusted standardized residuals (ASRs) for each cell. ASRs quantify the deviation between observed counts and the counts expected under a χ^2 test of independence. Cells with a |ASR| ≥1.96 were treated as statistically significant (approximately $p < 0.05$, two-tailed). Results are visualized as heatmaps in Fig. 12 (red = overrepresentation; blue = underrepresentation; shading indicates magnitude).

(a) RQ × Method. The RQ × M matrix indicates several patterned associations between research question types and methodological choices. RQ-F (teacher development/teacher knowledge) is overrepresented with M6 (ASR = +2.2), suggesting a synthesis-oriented orientation toward knowledge mapping. RQ-C (processes/mechanisms) shows an overrepresentation with M5 (ASR = +1.6), consistent with using structured tools, workflow modeling, or platforms to support intervention and process tracing. By contrast, RQ-D (assessment/measurement) is underrepresented with M2 (ASR = −2.8) and M3 (ASR = −1.8), indicating that assessment-focused work in this corpus relies less on interpretive or integrative designs. RQ-E is also underrepresented with M5 (ASR = −1.7). Taken together, these patterns are consistent with a corpus in which studies on levels and developmental trajectories often draw on existing instruments/tasks, whereas fewer studies engage in developing, calibrating, and validating context-specific tools or platforms. In that sense, links along a "develop–validate–apply" sequence appear uneven rather than systematically connected. Finally, the RQ-C × M8 cell (ASR = +4.4) likely reflects the "unspecified/other" coding category and residuals; we report it for completeness but do not interpret it substantively.

(b) RQ × Framework. The RQ × F matrix highlights several patterned associations between research-question types and framework choices. RQ-D (assessment/measurement) is associated with F5 (technology/tool-mediated frameworks) at a near-threshold level (ASR = +1.9), which is consistent with the use of structured tools and functional models to support operationalized, quantitatively oriented designs. RQ-B (design/implementation) shows a non-significant tendency to pair with F2 (inquiry/design frameworks) (ASR = +1.7), aligning with the DBR/DBIR/IBL traditions defined earlier. By contrast, RQ-C (processes/mechanisms)–F2 (ASR = −2.1) and RQ-B–F5 (ASR = −2.2) are underrepresented (observed less frequently than expected). In this dataset, these negative associations suggest that mechanism-focused questions are less often framed with inquiry/design frameworks, and design/implementation questions are less often framed through technology/tool mediation–areas that may warrant closer conceptual integration in future work. Finally, the RQ-C × F6 cell (ASR = +4.4) reflects unspecified/descriptive framing and is reported for completeness without substantive interpretation.

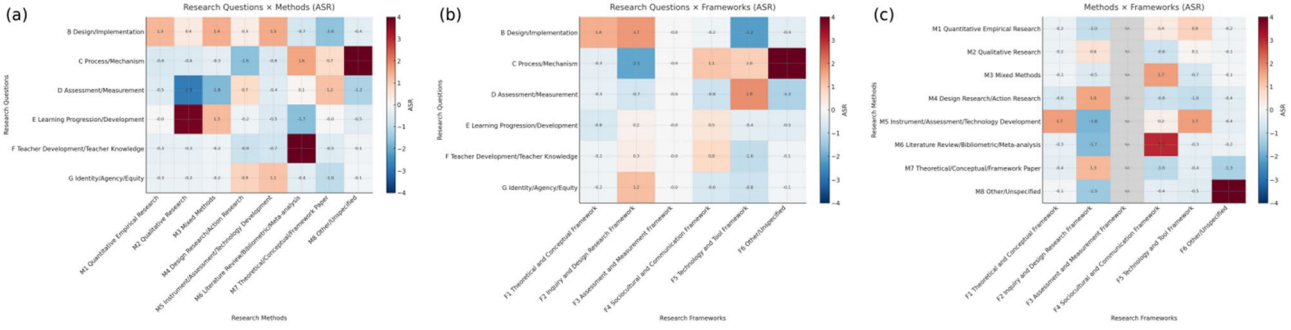


Fig. 12 Heatmap of ASRs for research question–method–framework combinations in scientific practice research in science education (2013–2023, WoS core collection)

(c) Method × Framework. The M × F matrix reveals several patterned associations, including one clearly overrepresented pair. M6 (reviews/bibliometric or meta-analytic studies) is overrepresented with F4 (sociocultural/communication frameworks) (ASR = +3.0), indicating that work oriented to culture, identity, and discourse is frequently consolidated through systematic syntheses and bibliometric mappings in this dataset. Several additional pairs show non-significant but consistently positive tendencies. M5 (instrument/assessment/technology development) tends to co-occur with F5 (technology/tool-mediated frameworks) and F1 (theoretical/conceptual frameworks) (both ASR = +1.7), consistent with tool/platform development that is grounded in organizing concepts and functional models. M4 (design-based/action research) shows a positive tendency to pair with F2 (inquiry/design frameworks) (ASR = +1.6), aligning with DBR/DBIR/IBL traditions defined earlier. M3 (mixed methods) also shows a positive tendency to pair with F4 (ASR = +1.7), consistent with integrating quantitative and qualitative evidence under sociocultural perspectives.In contrast, two pairings display nonsignificant underrepresentation–M5 × F2 (ASR = −1.8) and M6 × F2 (ASR = −1.7)–i.e., they were observed less frequently than expected. The former suggests weaker alignment in this dataset between tool/assessment development and inquiry/design frameworks (F2): relative to learning–design theories (DBR/IBL), M5-centered studies appear more often anchored in technology mediation or conceptual models (cf. the positive tendencies with F5/F1) than co-used with design frameworks. The latter indicates that review/bibliometric work is less frequently co-used with design frameworks, a pattern whose substantive meaning may require study-level inspection.

Discussion

Discussion of the results

From a publication-trend perspective, scholarship on scientific practices (SPs) shows a general increase over the study window, with faster growth in the later years and a local peak in 2021 (Fig. 2). Following the 2013 Next Generation Science Standards (NGSS), which explicitly elevated SPs to a core curricular goal, the literature has increasingly shifted from establishing why SPs matter to examining how they can be enacted in instruction. Against this backdrop, several distributional patterns are evident. By educational level, the corpus is concentrated in upper secondary (high school) and undergraduate settings, whereas early childhood contexts and preservice-teacher populations remain relatively sparse (Fig. 7). By discipline, research is dominated by integrated/general science and the core physical and biological sciences, with comparatively limited representation of geography, engineering, and technology and a relatively low share of cross-disciplinary integration (Fig. 8). At the co-authorship level, overall connectivity remains limited, although several small, thematically clustered collaboration groups can be identified (Fig. 3). Cross-national ties are anchored predominantly in Anglophone countries, and bridging nodes spanning languages and cultures appear comparatively rare. Taken together, these patterns suggest that cross-context comparability and generalizability merit further attention; to improve portability while maintaining contextual fit, further work may benefit from contextual calibration and greater inclusion of evidence from multilingual and non-core contexts.

Building on the macro-level patterns summarized above, we used CiteSpace keyword clustering and the timeline view to characterize what the field has been studying and how major topics connect to the classroom evidence chain. We then cross-checked these map-level observations against the observed coupling among research questions, methods, and theoretical/analytic frameworks (RQs–M–F). For readability, Table 7 condenses the key information from Figs. 4a, b and 6a—cluster proximity and links, timeline activity, and a brief, non-causal interpretation—into a compact reference.

To cross-check these map-level patterns against method-side evidence, we compiled ASR-based cross-tabulations linking themes with RQ–method–framework categories. Table 8 summarizes over and underrepresented pairings (by ASRs), thereby connecting the topical landscape to evidence production practices. Consistent with the ASR heatmaps (Fig. 12), assessment/measurement questions show a patterned association with tool/assessment/technology development methods and technology/tool-mediated frameworks and are less often paired with qualitative or mixed-methods designs. Process/mechanism questions also tend to co-occur with tool/assessment/technology development, while review/bibliometric work is overrepresented with sociocultural/communication frameworks. In line with the thematic maps, argumentation/explanation studies frequently rely on discourse/text analysis and rubric-based scoring; however, rubric dimensions and process indicators remain heterogeneous across studies, limiting cumulative comparability. Classroom/activity/engagement studies commonly align with sociocultural frameworks and, to a lesser extent, mixed-methods designs, yet their coupling to specific "evidence units" and "feedback pathways" appears comparatively weak in this corpus.

Taken together, the map-level observations (Table 7) and the coupling results (Table 8) highlight several

Table 7 Observable facts from the topic map and timeline

Theme/Main line	What the map shows (clusters & links)	What the timeline shows	Brief reading (no causal claims)
Evidence chain main line: #2 scientific argumentation; #3 scientific explanation; #4 assessing scientific practice	#2/#3 are adjacent and densely linked to #4 (see Fig 4a, b)	#2/#3 form sustained activity bands; #4 shows denser nodes in later years (see Fig. 6a)	Argumentation/explanation and assessment are advancing in parallel, but there is no clear sign of "confluence" yet.
Classroom ecology main line: #1 science classroom; #7 activity structure; #6 engaging student	#1/#7/#6 lie in a contiguous area on the map (see Fig 4a, b)	Long-running activity with steady growth after 2013 (see Fig. 6a)	Classroom organization/activity/participation constitute an "implementation–participation" region.
Methodological bridge: #5 action research	Located on the periphery yet connected to multiple clusters (see Fig 4a, b)	Activity is dispersed across time (see Fig. 6a)	Acts as a bridge for in-situ improvement, but signals of scaling remain limited.
Technology-related theme	No standalone cluster labeled "technology integration/learning analytics" (see Fig 4a, b)	—	Low salience on the map; likely embedded across other themes rather than consolidated as a single domain.

Note: The cluster map and the timeline are shown in Figs. 4a, b and 6a, respectively. This table reports only directly observable patterns from the figures and makes no causal claims

Table 8 Cross-evidence: Themes×RQ-M-F (based on ASRs)

Theme/Main line	RQ preferences (Fig. 12)	Method preferences (Fig. 12)	Framework preferences (Fig. 12)	Implications/Gap indicated
Assessment/measurement (#4)	RQ-D (assessment/measurement) overrepresented with F-5 (technology/tool-mediated) (ASR ≈ +1.9)	Tends to pair with M5 (tool/assessment/technology development); underrepresented with M2/M3 (ASR ≈ −2.8/-1.8)	F5 higher	Gap 1: cross-study misalignment of rubric dimensions and process indicators; limited interpretive/mixed evidence → use C–E–T/ECD to unify claim–evidence–task; report validity and cross-task comparability.
Argumentation/explanation (#2/#3)	RQs around evidence/reasoning quality and explanation quality	Frequent pairing of discourse/text analysis with rubric-based scoring	Often associated with CER/ECD	Advances in parallel with assessment but without convergence → strengthen formative uses and process evidence; use design-based/longitudinal studies to test the "task → process → feedback → learning" chain.
Classroom/activity/engagement (#1/#7/#6)	RQs on organization/participation/implementation	M3 (mixed methods) tends to be higher	F4 (sociocultural/communication) tends to be higher	Gap 2: weak evidence loop between implementation/participation and evidence quality → integrate activity sequences, evidence units, and feedback use.
Action research (#5)	—	M4 (design/action research) higher with F2 (inquiry/design) (ASR ≈ +1.6)	F2 higher	Gap 3: limits on scale and external validity → pursue cross-site/cross-context joint action research with shared task–rubric bundles.
Technology integration (dispersed)	RQ-E–M5 underrepresented (ASR ≈ −1.7)	M5–F5/F1 higher; M5–F2 underrepresented (ASR ≈ −1.8)	F5 and F1 higher; F2 lower	Gap 4: tech–pedagogy decoupling → move from substitution/augmentation to modification/redefinition; make technology endogenous to the evidence loop.

Note: The ASR heatmaps are shown in Fig. 12. Red = observed > expected; blue = observed < expected; |ASR| ≥1.96 is typically noteworthy (≈ $p < 0.05$). ASRs are exploratory and subject to multiple comparisons and sparse cells and should be interpreted with caution

structural weaknesses in this corpus. First, alignment shortfalls are evident: work on argumentation/explanation and assessment often develops in parallel with limited integration. Assessment studies also show concentrated preferences in methods and frameworks, and cross-study convergence in rubric dimensions, process indicators, and outcome measures remains limited; formative use and interpretive evidence are comparatively thin. Second, the evidence loop appears weak: the "implementation-participation" region (classroom organization, activity structure, student engagement) is adjacent to–but does not substantially overlap with–the "evidence-chain" region. This pattern suggests that activity sequences are designed primarily to support implementation and participation, whereas tighter alignment with "evidence units–feedback use–teacher decision-making"

is less consistently articulated; higher participation does not necessarily translate into higher-quality evidence use. Third, scalability and external validity remain constrained: action research connects classroom contexts with improvement cycles, yet it often takes the form of dispersed, small-sample, context-bound cases. The limited use of shared rubrics and task bundles further constrains cross-site comparability and portability. Fourth, a technology–pedagogy decoupling is suggested: the maps lack a salient "technology integration" cluster, and the ASR pattern indicates that technology development is more often anchored in functional or conceptual models than in inquiry/design frameworks. Together, these cues imply that technology has not yet become an endogenous component of the classroom evidence loop.

In response, we propose four coordinated directions: alignment, closure, scale, and technological endogenization. (1) Evidence-centered, alignment-oriented design (e.g., claim–evidence–task mapping and constructive alignment) can be adopted upfront to better standardize rubric dimensions and process indicators across studies, to report validity evidence and cross-task comparability more systematically, and to embed formative use within task sequences. (2) Loop-closure through integrated design should be strengthened across activity sequences, evidence units, and feedback use, so that the "data → feedback → teacher decision-making" pathway becomes a routine feature of classroom practice and the "implementation–participation" and "evidence–chain" regions are more tightly connected. (3) Scalable validation can be advanced through cross-site and cross-cultural joint action research using shared task-rubric bundles, while also expanding evidence from younger learners, preservice cohorts, and non-Anglophone contexts to improve external validity and contextual fit. (4) Technology-enabled endogenization requires moving beyond substitution/augmentation toward modification/redefinition, leveraging platform logs and learning analytics to link process data, feedback, and teacher decisions, and thereby positioning technology as an endogenous element of the evidence ecology rather than an external add-on.

Implications

Methodological implications

Building on standard bibliometric visualization, this study integrates CiteSpace-based keyword co-occurrence clustering and timeline analysis with ASR heatmaps derived from three contingency tables (RQ × M, RQ × F, and M × F), thereby creating an analytic pipeline from thematic cartography to structural diagnosis. The mapping component traces major themes and their temporal trajectories. The diagnostic component uses adjusted standardized residuals (ASR; $|ASR| \geq 1.96$ as a salient threshold) to make over- and underrepresented pairings explicit, shifting bibliometric reviews from cataloguing hotspots to diagnosing alignment patterns. In this way, the approach moves beyond identifying where topics cluster to examining how systematic preferences or gaps arise across specific RQ–method–framework linkages, offering actionable evidence for research design and agenda setting.

Cartographic implications

Using keyword co-occurrence clustering and timeline analysis, the knowledge maps provide an integrated view of the thematic structure and temporal evolution of research on scientific practices (SPs). On the one hand, the maps highlight core topics and–drawing on node weights and connectivity–surface representative authors and institutions associated with these topics. On the other hand, the timeline view clarifies how attention shifts from conceptual framing to classroom enactment, assessment, and technology, and, together with indicators such as cluster size, centrality, and temporal persistence, helps identify lines of work that remain active. Accordingly, the maps move beyond a descriptive snapshot and offer an evidence-informed reference for identifying and prioritizing research directions.

Substantive implications

In summary, the structural diagnosis suggests that research on scientific practices (SPs) is moving from hotspot accumulation toward system building. At the level of populations and settings, the maps indicate a concentration in upper secondary and undergraduate contexts, with collaborations anchored largely in Anglophone settings. This pattern points to the need to broaden coverage to early grades and preservice teachers, informal learning environments, and cross-cultural settings, while also adopting open task- and data-sharing mechanisms to strengthen external validity and cross-context comparability. At the level of mechanism building, outcome reporting appears to outpace explanatory work. Future studies may therefore benefit from designs that support causal and process inferences, such as incorporating controls and delayed posttests, collecting process-tracing data (e.g., learning trajectories, interaction logs), and pairing these with mediation/moderation analyses and mixed-effects models to clarify the intervention–process–capability pathway. With respect to design coherence, assessment and instruction are often positioned side by side rather than designed to be isomorphic. This suggests prioritizing goals–tasks–assessment alignment and more explicitly integrating identity, agency, and equity variables into intervention designs, ideally in connection with teacher professional learning, to better close the theory–tool–practice loop. Finally,

within the evidence system, rapid tool proliferation coexists with limited cross-study accumulation. This tension highlights the value of advancing anchor tasks, scoring protocols, and rater training, together with validity chains (content–structural–external) and clear rules for metadata archiving and reuse, to build a comparable and integrable evidence platform. Collectively, these implications translate the structural findings into coordinated lines of action across four dimensions–populations/settings, mechanisms, design, and evidence–supporting mechanism-driven and cumulative research across contexts.

Limitations

Although this review provides a systematic picture of the knowledge structure and developmental features of research on scientific practices (SPs), several limitations should be noted. First, the corpus is limited to English-language journal articles indexed in the Web of Science Core Collection (WoSCC) over 2013–2023. Relevant work published in other languages, in non-journal venues (e.g., books, chapters, conference proceedings), or indexed outside WoSCC may therefore be underrepresented, and the dataset may also be affected by indexing lag. In addition, our search was centered on the term "scientific practice*", which may under-capture practice-adjacent studies framed with alternative terminology (e.g., NGSS "Science and Engineering Practices", scientific inquiry, engineering practices). This term choice reflects our intent to map scholarship that explicitly uses "scientific practice(s)" as its framing construct; accordingly, the present review should be read as a focused, term-explicit mapping within the WoSCC "Education & Educational Research" corpus, rather than as an exhaustive synthesis of all SEP- or practice-adjacent literatures. Relatedly, because retrieval relied mainly on the WoS category filter, relevant records classified in adjacent categories may have been missed. Future work could broaden the term set (e.g., adding "science education") and expand category coverage to test the robustness of the mapped patterns under alternative retrieval specifications; thus, the findings should be interpreted as corpus-conditional with respect to generalizability. Second, the classification of research questions, methods, and theoretical frameworks relies primarily on manual coding. Although we used a clear codebook and cross-checking procedures, some subjectivity cannot be entirely ruled out. Moreover, the ASR heatmap diagnostics depend on coding granularity and threshold choices (e.g., $|ASR| \geq 1.96$), and may be sensitive when contingency tables are sparse. Third, as with most bibliometric mapping, co-occurrence clustering and related visualizations can be sensitive to preprocessing and parameter settings (e.g., keyword normalization and clustering/threshold choices), and co-citation patterns partly reflect citation practices and time-lag effects. Fourth, while we triangulated keyword co-occurrence with co-citation analysis and burst detection to better characterize knowledge bases and emerging fronts, we did not conduct study-level quality appraisal or meta-analytic synthesis; therefore, evidence-strength weighting and effect-size aggregation remain beyond the scope of this review. Finally, the review window ends in mid-2023, and very recent developments (e.g., generative-AI-supported instruction and immersive laboratory environments) are not captured.

Conclusions

Drawing on 263 articles indexed in the Web of Science Core Collection, this review integrates bibliometric mapping with content analysis to characterize the knowledge structure and developmental features of research on scientific practices in science education, within the defined corpus across five lenses: publication trends, study populations, international collaboration, thematic structure, and methodological/theoretical frameworks. The results suggest that, at the macro level, this literature can be read as reflecting a broad trajectory—from conceptual framing, to instructional enactment, to learning assessment—alongside emerging linkages among research questions, methods, and theoretical/analytical frameworks. Substantively, the mapped work concentrates on the teacher–student–task triad, with classroom enactment studies and assessment-instrument development often advancing in parallel. At the same time, the synthesis highlights several recurring gaps, including limitation in representativeness and generalizability, a relative shortage of mechanism-oriented designs, fragmented connections among instruction, assessment, and technology, insufficient attention to sociocultural dimensions, and underdeveloped approaches for assessing higher-order practices. Accordingly, we outline five directions for future work: (1) broaden representativeness; (2) strengthen explanatory accounts using process evidence (including causal and process-oriented designs where feasible); (3) promote coherent alignment among theory, tools, and practice; (4) more fully incorporate sociocultural dimensions; and (5) develop and validate more robust and comparable approaches for higher-order assessment. Progress along these lines may support a shift in the field from hotspot aggregation toward cumulative system building, and from outcome reporting toward mechanism-focused explanation.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s43031-026-00167-x.

Supplementary Material 1

Supplementary Material 2

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Author contributions
Conceptualization, Y.L.C.and L. W.; methodology,Y.L.C. and L. W.; validation, Y.L.C. and L. W.; formal analysis, Y.L.C.and X.J.L.; writing original draft,Y.L.C.; writing—review and editing, Y.L.C.,X.J.W., and L. W.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Competing interests
The authors declare no competing interests.

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References

Al-Gaseem, M. M., & Khasawneh, M. A. S. (2022). Action research in science teacher education program: Significance and benefit according to the students-teachers' assessments. *Przestrzeń Społeczna (Social Space)*, *22*(3), 97–113. https://socialspacejournal.eu/menu-script/index.php/ssj/article/view/119

Çalık, M., & Sözbilir, M. (2014). İçerik Analizinin Parametreleri. *Education and Science*, *39*(174), 33–38. https://doi.org/10.15390/EB.2014.3412

Al, U., & Tonta, Y. (2004). Atıf analizi: Hacettepe Üniversitesi Kütüphanecilik Bölümü tezlerinde atıf yapılan kaynaklar [Citation analysis: Sources cited in dissertations completed at Hacettepe University Department of librarianship]. *Bilgi Dünyası*, *5*(1), 19–47. https://doi.org/10.15612/BD.2004.497

Australian CurriculumAustralian Curriculum, Assessment and Reporting Authority. (n.d.). Structure (the Australian curriculum: Science, version 8.4). January 7, 2026, from https://v8.australiancurriculum.edu.au/f-10-curriculum/science/structure/

Avraamidou, L. (2020). Science identity as a landscape of becoming: Rethinking recognition and emotions through an intersectionality lens. *Cultural Studies of Science Education*, *15*(2), 323–345. https://doi.org/10.1007/s11422-019-09954-7

Bauer, M. W. (2000). Classical content analysis: A review. In M. W. Bauer, & G. Gaskell (Eds.), *Qualitative researching with text, image and sound: A practical handbook for social research* (pp. 131–151). SAGE. https://doi.org/10.4135/9781849209731

Benedict-Chambers, A., Kademian, S. M., Davis, E. A., & Palincsar, A. S. (2017). Guiding students towards sensemaking: Teacher questions focused on integrating scientific practices with science content. *International Journal of Science Education*, *39*(15), 1977–2001. https://doi.org/10.1080/09500693.2017.1366674

Bennion, A. H., & Davis, E. A. (2024). Characterizing engagement in the science practices: A study of preservice elementary teachers. *Science Education*, *108*(5), 1392–1419. https://doi.org/10.1002/sce.21876

Berland, L. K., & McNeill, K. L. (2010). A learning progression for scientific argumentation: Understanding student work and designing supportive instructional contexts. *Science Education*, *94*(5), 765–793. https://doi.org/10.1002/sce.20402

Berland, L. K., & Reiser, B. J. (2009). Making sense of argumentation and explanation. *Science Education*, *93*(1), 26–55. https://doi.org/10.1002/sce.20286

Brandes, U. (2001). A faster algorithm for betweenness centrality. *Journal of Mathematical Sociology*, *25*(2), 163–177. https://doi.org/10.1080/0022250X.2001.9990249

Bybee, R. W. (2014). *The case for STEM education: Challenges and opportunities*. NSTA Press

Cardace, A., Wilson, M., & Metz, K. E. (2021). Designing a learning progression about micro-evolution to inform instruction and assessment in elementary science. *Education Sciences*, *11*(10), 609. https://doi.org/10.3390/educsci11100609

Chan, P., Van Gerven, T., Dubois, J.-L., & Bernaerts, K. (2021). Virtual chemical laboratories: A systematic literature review of research, technologies and instructional design. *Computers and Education Open*, *2*, 100053. https://doi.org/10.1016/j.caeo.2021.100053

Chen, C. (2006). CiteSpace II: Detecting and visualizing emerging trends and transient patterns in scientific literature. *Journal of the American Society for Information Science and Technology*, *57*(3), 359–377. https://doi.org/10.1002/asi.20317

Chen, C., Ibekwe-SanJuan, F., & Hou, J. (2010). The structure and dynamics of cocitation clusters: A multiple-perspective cocitation analysis. *Journal of the American Society for Information Science and Technology*, *61*(7), 1386–1409. https://doi.org/10.1002/asi.21309

Cochran-Smith, M., & Lytle, S. L. (2009). *Inquiry as stance: Practitioner research for the next generation*. Teachers College Press

Cole, M. (1996). *Cultural psychology: A once and future discipline*. Harvard University Press

Cooper, A. C., Southard, K. M., Osness, J. B., & Bolger, M. S. (2022). The instructor's role in a model-based inquiry laboratory: Characterizing instructor supports and intentions in teaching authentic scientific practices. *CBE—Life Sciences Education*, *21*(1), ar9. https://doi.org/10.1187/cbe.21-07-0177

Costa, J., Leite, L., & Niccolai, G. (2021). A framework for understanding the role of scientific practices in science education. *Research in Science Education*, *50*(2), 397–427. https://doi.org/10.1007/s11165-018-9703-2

Costa, S. L. R., & Broietti, F. C. D. (2021). Scientific practices in science education publications: An analysis of research contexts. *Science Education International*, *32*(4), 282–291. https://doi.org/10.33828/sei.v32.i4.1

Creswell, J. W. (2012). *Educational research: Planning, conducting, and evaluating quantitative and qualitative research* (4th ed.). Pearson

Criswell, B. A., & Rushton, G. T. (2014). Activity structures and the unfolding of problem-solving actions in high-school chemistry classrooms. *Research in Science Education*, *44*(1), 155–188. https://doi.org/10.1007/s11165-013-9374-x

Department for Education. (2013, September 11). *National curriculum in England: Science programmes of study*. GOV.UK. https://www.gov.uk/government/publications/national-curriculum-in-england-science-programmes-of-study

Dinçer, S. (2018). Eğitim Bilimleri Araştırmalarında İçerik Analizi: Meta-Analiz, Meta-Sentez, Betimsel İçerik Analizi. *Bartın Üniversitesi Eğitim Fakültesi Dergisi*, *7*(1), 176–190. https://doi.org/10.14686/buefad.363159

Evagorou, M., Nicolaou, C., & Lymbouridou, C. (2020). Modelling and argumentation with elementary school students. *Canadian Journal of Science, Mathematics and Technology Education*, *20*(1), 58–73. https://doi.org/10.1007/s42330-020-00076-9

Finnish National Agency for Education. (n.d.). National core curriculum for primary and lower secondary (basic) education. January 7, 2026, from https://www.oph.fi/en/education-and-qualifications/national-core-curriculum-primary-and-lower-secondary-basic-education

FinnishNational Board of Education. (2014). *National core curriculum for basic education 2014* (Perusopetuksen opetussuunnitelman perusteet 2014). Finnish National Board of Education. https://www.oph.fi/en/statistics-and-publications/publications/new-national-core-curriculum-basic-education-focus-school

Ford, M. J., & Forman, E. A. (2006). Chapter 1: Redefining disciplinary learning in classroom contexts. *Review of Research in Education*, *30*(1), 1–32. https://doi.org/10.3102/0091732X030001001

Forman, E. A. (2018). The practice turn in learning theory and science education. In D. W. Kritt (Ed.), *Constructivist education in an age of accountability* (pp. 97–104). Palgrave Macmillan

Freeman, L. C. (1977). A set of measures of centrality based on betweenness. *Sociometry*, *40*(1), 35–41. https://doi.org/10.2307/3033543

Glänzel, W., & Schubert, A. (2004). Analysing scientific networks through co-authorship. In H. F. Moed, W. Glänzel, & U. Schmoch (Eds.), *Handbook of quantitative science and technology research: The use of publication and patent statistics in studies of S & T systems* (pp. 257–276). https://doi.org/10.1007/1-4020-2755-9_12

González-Howard, M., & McNeill, K. L. (2019). Teachers' framing of argumentation goals: Working together to develop individual versus communal

understanding. *Journal of Research in Science Teaching*, *56*(6), 821–844. https://doi.org/10.1002/tea.21530
Gough, D., Oliver, S., & Thomas, J. (2012). *An introduction to systematic reviews*. SAGE
Hao, J., & Mislevy, R. J. (2018). The evidence trace file: A data structure for virtual performance assessments informed by data analytics and evidence-centered design. *ETS Research Report Series*, *2018*(1), 1–16. https://doi.org/10.1002/ets2.12215
Hendratmoko,A. F., Madlazim, M., Widodo, W., Suyono, S., & Supardi, Z. A. I. (2024). Inquiry and debate in science learning: Potential strategy for improving students' scientific argumentation skills. *International Journal of Education in Mathematics, Science and Technology, 12*(1), 114–138. https://doi.org/10.46328/ijemst.3152
Hosbein, K. N., Alvarez-Bell, R., Callis-Duehl, K. L., Sampson, V., Wolf, S. F., & Walker, J. P. (2021). Development of the investigation design, explanation, and argument assessment for general chemistry I laboratory. *Journal of Chemical Education*, *98*(2), 293–306. https://doi.org/10.1021/acs.jchemed.0c01075
Çiltaş, A., Güler, G., & Sözbilir, M. (2012). Mathematics education research in Turkey: A content analysis study. *Educational Sciences: Theory & Practice*, *12*(1), 574–578
Jian, M., Jin, D., & Wu, X. (2023). Research hotspots and development trends of international learning cycle model: Bibliometric analysis based on CiteSpace. *Heliyon*, *9*(11), e22076. https://doi.org/10.1016/j.heliyon.2023.e22076
Jin, D., & Jian, M. (2024). Research hotspots and development trends of model and modelling education research: Bibliometric analysis based on CiteSpace. *Heliyon*, *10*(11), e32590. https://doi.org/10.1016/j.heliyon.2024.e32590
Jin, H., & Anderson, C. W. (2012). A learning progression for energy in socio-ecological systems. *Journal of Research in Science Teaching*, *49*(9), 1149–1180. https://doi.org/10.1002/tea.21051
Kelly, G. J. (2014). The social bases of disciplinary knowledge and practice in productive disciplinary engagement. *International Journal of Educational Research*, *64*, 211–214. https://doi.org/10.1016/j.ijer.2013.07.008
Kemmis, S., & McTaggart, R. (1988). *The action research planner* (3rd ed.). Deakin University Press
Killpack, T. L., & Popolizio, T. R. (2023). An equity-focused redesign of an introductory organismal biology lab course to develop foundational scientific practices. *Journal of Microbiology & Biology Education*, *24*(1), e00213–22. https://doi.org/10.1128/jmbe.00213-22
Koomen, M. H., Rodriguez, E., Hoffman, A., Petersen, C., & Oberhauser, K. (2018). Authentic science with citizen science and student-driven science fair projects. *Science Education*, *102*(3), 593–644. https://doi.org/10.1002/sce.21335
Krajcik, J. S., McNeill, K. L., & Reiser, B. J. (2008). Learning-goals-driven design model: Developing curriculum materials that align with national standards and incorporate project-based pedagogy. *Science Education*, *92*(1), 1–32. https://doi.org/10.1002/sce.20240
Laverty, J. T., Underwood, S. M., Matz, R. L., Posey, L. A., Carmel, J. H., Caballero, M. D., & Jardeleza, S. E. (2016). Characterizing college science assessments: The three-dimensional learning assessment protocol. *PLoS One*, *11*(9), e0162333. https://doi.org/10.1371/journal.pone.0162333
Lee, M.-H., Wu, Y.-T., & Tsai, C.-C. (2009). Research trends in science education from 2003 to 2007: A content analysis of publications in selected journals. *International Journal of Science Education*, *31*(15), 1999–2020. https://doi.org/10.1080/09500690802314876
Leowardy, M., Lee, Y.-J., & Ong, Y. S. (2025). A decade of scientific and engineering practices: A scoping review. *Studies in Science Education*, 1–45. Advance online publication. https://doi.org/10.1080/03057267.2025.2481356
Li, K., Rollins, J., & Yan, E. (2018). Web of science use in published research and review papers 1997–2017: A selective, dynamic, cross-domain, content-based analysis. *Scientometrics*, *115*(1), 1–20. https://doi.org/10.1007/s11192-017-2622-5
Martin, J. D., & Graulich, N. (2023). When a machine detects student reasoning: A review of machine learning-based formative assessment of mechanistic reasoning. *Chemistry Education Research and Practice*, *24*(2), 407–427. https://doi.org/10.1039/D2RP00287F
McNeill, K. L., & Knight, A. M. (2013). Teachers' pedagogical content knowledge of scientific argumentation: The impact of professional development on K-12 teachers. *Science Education*, *97*(6), 936–972. https://doi.org/10.1002/sce.21081
Miller, E., Manz, E., Russ, R. S., Stroupe, D., & Berland, L. K. (2018). Addressing the epistemic elephant in the room: Epistemic agency and the Next Generation science Standards. *Journal of Research in Science Teaching*, *55*(7), 1053–1075. https://doi.org/10.1002/tea.21459
Ministry of Education of the People's Republic of China. (2022, April 20). 教育部关于印发义务教育课程方案和课程标准 （2022年版） 的通知. https://www.moe.gov.cn/srcsite/A26/s8001/202204/t20220420_619921.html
Ministry of Education, Singapore. (2013). Science syllabus: Primary (implementation starting with 2014 primary three cohort). *Curriculum planning and development division*. https://www.moe.gov.sg/-/media/files/primary/science-primary-2014.pdf
Moed, H. F. (2005). *Citation analysis in research evaluation*. Springer. https://doi.org/10.1007/1-4020-3714-7
Mongeon, P., & Paul-Hus, A. (2016). The journal coverage of Web of science and scopus: A comparative analysis. *Scientometrics*, *106*(1), 213–228. https://doi.org/10.1007/s11192-015-1765-5
National Research Council. (2012). *A framework for K-12 science education: Practices, crosscutting concepts, and core ideas*. National Academies Press. https://doi.org/10.17226/13165
Newman, M. E. J. (2001). The structure of scientific collaboration networks. *Proceedings of the National Academy of Sciences of the United States of America*, *98*(2), 404–409. https://doi.org/10.1073/pnas.98.2.404
Newman, M. E. J. (2004). Coauthorship networks and patterns of scientific collaboration. *Proceedings of the National Academy of Sciences of the United States of America*, *101*(Suppl. 1), 5200–5205. https://doi.org/10.1073/pnas.0307545100
NGSS Lead States. (2013). *Next Generation science Standards: For states, by states*. National Academies Press
Nicolaou, C. T., & Constantinou, C. P. (2014). Assessment of the modeling competence: A systematic review and synthesis of empirical research. *Educational Research Review*, *13*, 52–73. https://doi.org/10.1016/j.edurev.2014.10.001
Oh, P. S., Ha, H., & Yoo, Y. J. (2022). Epistemological messages in a modeling-based elementary science classroom compared with a traditional classroom. *Science Education*, *106*(4), 797–829. https://doi.org/10.1002/sce.21724
Osborne, J. (2010). Arguing to learn in science: The role of collaborative, critical discourse. *Science*, *328*(5977), 463–466. https://doi.org/10.1126/science.1183944
Osborne, J., Erduran, S., & Simon, S. (2004). Enhancing the quality of argumentation in school science. *Journal of Research in Science Teaching*, *41*(10), 994–1020. https://doi.org/10.1002/tea.20035
Petticrew, M., & Roberts, H. (2006). *Systematic reviews in the social sciences: A practical guide*. Blackwell Publishing. https://doi.org/10.1002/9780470754887
Pinto, M., Fernández-Pascual, R., Caballero-Mariscal, D., Sales, D., Guerrero, D., & Uribe, A. (2019). Scientific production on mobile information literacy in higher education: A bibliometric analysis (2006–2017). *Scientometrics*, *120*(1), 57–85. https://doi.org/10.1007/s11192-019-03115-x
Pizzolato, L. A. V., & Tsuji, L. J. S. (2022). Citizen science in K–12 school-based learning settings. *School Science and Mathematics*, *122*(4), 222–231. https://doi.org/10.1111/ssm.12528
Prins, G. T., Bulte, A. M. W., & Pilot, A. (2018). Designing context-based teaching materials by transforming authentic scientific modelling practices in chemistry. *International Journal of Science Education*, *40*(10), 1108–1135. https://doi.org/10.1080/09500693.2018.1470347
Rönnebeck, S., Bernholt, S., & Ropohl, M. (2016). Searching for a common ground – a literature review of empirical research on scientific inquiry activities. *Studies in Science Education*, *52*(2), 161–197. https://doi.org/10.1080/03057267.2016.1206351
Saribas, D. (2022). Action research on preservice elementary teachers' understandings of the scientific method and the use of evidence in a science and technology teaching course. *International Journal of Science Education*, *44*(2), 277–298. https://doi.org/10.1080/02635143.2020.1814233
Saribas, D., & Akdemir, Z. G. (2022). Action research on pre-service elementary teachers' understandings of the scientific method and the use of evidence in a science and technology teaching course. *Research in Science & Technological Education*, *40*(4), 431–453. https://doi.org/10.1080/02635143.2020.1814233
Sherrer, S. M. (2020). A virtual laboratory module exploring photosynthesis during COVID-19. *Biochemistry and Molecular Biology Education*, *48*(6), 659–661. https://doi.org/10.1002/bmb.21464
Stroupe, D. (2014). Examining classroom science practice communities: How teachers and students negotiate epistemic agency and learn science-as-practice. *Science Education*, *98*(3), 487–516. https://doi.org/10.1002/sce.21112
Sulaiman, N., Werth, A., & Lewandowski, H. J. (2023). Students' views about experimental physics in a large-enrollment introductory lab focused on experimental scientific practices. *Physical Review Physics Education Research*, *19*(1), 010116. https://doi.org/10.1103/PhysRevPhysEducRes.19.010116
Sun,L., Guo, Z., & Hu, L. (2021). Educational games promote the development of students' computational thinking: A meta-analytic review. *Interactive Learning Environments*. https://doi.org/10.1080/10494820.2021.1931891

Tho, S. W., Yeung, Y. Y., Wei, R., Chan, K. W., & So, W. W. (2017). A systematic review of remote laboratory work in science education with the support of visualizing its structure through the HistCite and CiteSpace software. *International Journal of Science & Mathematics Education*, *15*(7), 1217–1236. https://doi.org/10.1007/s10763-016-9740-z

Tikva,C., & Tambouris, E. (2021). Mapping computational thinking through programming in K-12 education: A conceptual model based on a systematic literature review. *Computers & Education, 162*, Article 104083. https://doi.org/10.1016/j.compedu.2020.104083

Tsai, C.-C., & Wen, M. L. (2005). Research and trends in science education from 1998 to 2002: A content analysis of publication in selected journals. *International Journal of Science Education*, *27*(1), 3–14. https://doi.org/10.1080/09500690420002437 27

Underwood, S. M., Posey, L. A., Herrington, D. G., Carmel, J. H., & Cooper, M. M. (2018). Adapting assessment tasks to support three-dimensional learning. *Journal of Chemical Education*, *95*(2), 207–217. https://doi.org/10.1021/acs.jchemed.7b00645

Vamvakas, M., Tytler, R., & White, P. J. (2023). Translating contemporary scientists' knowledge and practice into classrooms: Scalable design supporting identity work. *Frontiers in Education*, *8*, 1100171. https://doi.org/10.3389/feduc.2023.1100171

Wagner, C. S., & Leydesdorff, L. (2005). Network structure, self-organization, and the growth of international collaboration in science. *Research Policy*, *34*(10), 1608–1618. https://doi.org/10.1016/j.respol.2005.08.002

Walsh, D., & Downe, S. (2005). Meta-synthesis method for qualitative research: A literature review. *Journal of Advanced Nursing*, *50*(2), 204–211. https://doi.org/10.1111/j.1365-2648.2005.03380.x

Waltman, L. (2016). A review of the literature on citation impact indicators. *Journal of Informetrics*, *10*(2), 365–391. https://doi.org/10.1016/j.joi.2016.02.007

Wang, D., Song, C., & Barabási, A.-L. (2013). Quantifying long-term scientific impact. *Science*, *342*(6154), 127–132. https://doi.org/10.1126/science.1237825

Wang, S., Chen, Y., Lv, X., & Xu, J. (2023). Hot topics and frontier evolution of science education research: A bibliometric mapping from 2001 to 2020. *Science & Education*, *32*(3), 845–869. https://doi.org/10.1007/s11191-022-00337-z

Wyner, J., Doherty, J. H., & Kahn, J., et al. (2020). Developing a learning progression for three-dimensional learning of the patterns of evolution. *Science Education*, *104*(6), 1072–1102. https://doi.org/10.1002/sce.21597

Yuriev, E., Naidu, S., Schembri, L. S., & Short, J. L. (2017). Scaffolding the development of problem-solving skills in chemistry: Guiding novice students out of dead ends and false starts. *Chemistry Education Research and Practice*, *18*(3), 486–504. https://doi.org/10.1039/C7RP00009J

Zupic, I., & Čater, T. (2015). Bibliometric methods in management and organization. *Organizational Research Methods*, *18*(3), 429–472. https://doi.org/10.1177/1094428114562629

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